Annual Report - General Order No. R5-2007-0035
Reporting period 01/01/2023 to 12/31/2023.

DAIRY FACILITY INFORMATION

A. NAME OF DAIRY OR BUSINESS OPERATING THE DAIRY: Rocking S Dairy #1

Physical address of dairy:

Number and Street	City	County	Zip Code
3301 Ladd RD	Modesto	Stanislaus	95356

Street and nearest cross street (if no address):

Date facility was originally placed in operation: 12/07/1979

Regional Water Quality Control Board Basin Plan designation: San Joaquin River Basin

County Assessor Parcel Number(s) for dairy facility:

X003-X024-X006-XXXX X030-X240-X070-XXXX

B. OPERATORS

teVelde, George

Operator name: teVelde, George

Telephone no.: (209) 000-0000

Landline	Cellular

Mailing Address Number and Street	City	State	Zip Code
3301 Ladd RD	Modesto	CA	95356

This operator is responsible for paying permit fees.

C. OWNERS

teVelde, George

Legal owner name: teVelde, George

Telephone no.: (209) 000-0000

Landline	Cellular

Mailing Address Number and Street	City	State	Zip Code
3301 Ladd RD	Modesto	CA	95356

This owner is responsible for paying permit fees.

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AVAILABLE NUTRIENTS

A. HERD INFORMATION

	Milk Cows	Dry Cows	Bred Heifers (15-24 mo.)	Heifers (7-14 mo. to breeding)	Calves (4-6 mo.)	Calves (0-3 mo.)
Number open confinement	0	0	500	1,150	0	0
Number under roof	1,250	0	0	0	0	0
Maximum number	1,280	0	550	1,200	0	0
Average number	1,250	0	500	1,150	0	0
Avg live weight (lbs)	1,350	0	1,050	600		

Predominant milk cow breed: Holstein

Average milk production: 80 *pounds per cow per day*

B. MANURE GENERATED

Total manure excreted by the herd: 49,359.73 *tons per reporting period*

Total nitrogen from manure: 594,990.87 *lbs per reporting period*

After ammonia losses (30% loss applied): 416,493.61 *lbs per reporting period*

Total phosphorus from manure: 100,998.11 *lbs per reporting period*

Total potassium from manure: 247,599.03 *lbs per reporting period*

Total salt from manure: 588,562.50 *lbs per reporting period*

C. PROCESS WASTEWATER GENERATED

Process wastewater generated: 65,963,640 *gallons*

Total nitrogen generated: 79,274.99 *lbs*

Total phosphorus generated: 22,529.00 *lbs*

Total potassium generated: 109,633.17 *lbs*

Total salt generated: 457,853.93 *lbs*

	65,963,640	*gallons applied*
+	0	*gallons exported*
-	0	*gallons imported*
=	65,963,640	*gallons generated*

D. FRESH WATER SOURCES

Source Description	Type
Ag Well #1	Ground water
Ag Well #3	Ground water
Ag Well #4	Ground water
Ag Well #5	Ground water
Canal Water	Surface water

Source Description	Type
Domestic Well	Ground water

E. SUBSURFACE (TILE) DRAINAGE SOURCES

No subsurface (tile) drainage sources entered.

F. NUTRIENT IMPORTS

No dry manure nutrient imports entered.

No process wastewater nutrient imports entered.

No commercial or other nutrient imports entered.

G. NUTRIENT EXPORTS

No solid nutrient exports entered.

No liquid nutrient exports entered.

APPLICATION AREA

A. LIST OF LAND APPLICATION AREAS

Field name	Controlled acres	Cropable acres	Total harvests	Type of waste applied	Parcel number
1	33	33	1	process wastewater	X003-X024-X009-XXXX
1W	17	17	0	none	X003-X024-X009-XXXX
2A	39	39	1	process wastewater	X003-X024-X009-XXXX
2B	5	5	1	process wastewater	X003-X024-X009-XXXX
2W	38	38	0	none	X003-X024-X013-XXXX
3	33	33	2	process wastewater	X003-X024-X009-XXXX
3W	20	20	0	none	X003-X024-X013-XXXX
4	36	36	2	process wastewater	X003-X024-X009-XXXX
4W	16	16	0	none	X003-X024-X012-XXXX
5A	14	14	2	process wastewater	X003-X024-X009-XXXX
5B	12	12	2	process wastewater	X003-X024-X009-XXXX
5W	20	20	0	none	X003-X024-X013-XXXX
6A	58	58	2	process wastewater	X003-X024-X008-XXXX
6B	7	7	2	process wastewater	X003-X024-X008-XXXX
7	31	31	2	process wastewater	X003-X024-X013-XXXX
8	14	14	2	process wastewater	X003-X024-X012-XXXX
9	18	18	2	process wastewater	X003-X024-X013-XXXX
Totals for areas that were used for application	300	300	21		
Totals for areas that were not used for application	111	111	0		
Land application area totals	411	411	21		

B. CROPS AND HARVESTS

1

Field name: 1

1

05/15/2023: Corn, silage

Crop: Corn, silage Acres planted: 33 Plant date: 05/15/2023

Harvest date	Yield	Reporting basis	Density (lbs/cu ft)	Moisture (%)	N (mg/kg)	P (mg/kg)	K (mg/kg)	Salt (mg/kg)	TFS (%)
09/15/2023	1,089.00 *ton*	Dry-weight		68.3	17,600.00	2,140.00	8,790.00		5.62

	Yield (tons/acre)	Total N (lbs/acre)	Total P (lbs/acre)	Total K (lbs/acre)	Salt (lbs/acre)
Anticipated harvest content	30.00	240.00	45.00	198.00	0.00
Total actual harvest content	33.00	368.23	44.77	183.90	1,175.82

2A

Field name: 2A

05/15/2023: Corn, silage

Crop: Corn, silage Acres planted: 39 Plant date: 05/15/2023

Harvest date	Yield	Reporting basis	Density (lbs/cu ft)	Moisture (%)	N (mg/kg)	P (mg/kg)	K (mg/kg)	Salt (mg/kg)	TFS (%)
09/15/2023	1,326.00 *ton*	Dry-weight		70.1	16,200.00	1,740.00	9,380.00		5.68

	Yield (tons/acre)	Total N (lbs/acre)	Total P (lbs/acre)	Total K (lbs/acre)	Salt (lbs/acre)
Anticipated harvest content	30.00	240.00	45.00	198.00	0.00
Total actual harvest content	34.00	329.38	35.38	190.71	1,154.86

2B

Field name: 2B

2B

05/15/2023: Corn, silage

Crop: Corn, silage Acres planted: 5 Plant date: 05/15/2023

Harvest date	Yield	Reporting basis	Density (lbs/cu ft)	Moisture (%)	N (mg/kg)	P (mg/kg)	K (mg/kg)	Salt (mg/kg)	TFS (%)
09/16/2023	170.00 *ton*	Dry-weight		70.1	16,200.00	1,740.00	9,380.00		5.68

	Yield (tons/acre)	Total N (lbs/acre)	Total P (lbs/acre)	Total K (lbs/acre)	Salt (lbs/acre)
Anticipated harvest content	30.00	240.00	45.00	198.00	0.00
Total actual harvest content	34.00	329.38	35.38	190.71	1,154.86

3

Field name: 3

11/01/2022: Oats, silage-soft dough

Crop: Oats, silage-soft dough Acres planted: 33 Plant date: 11/01/2022

Harvest date	Yield	Reporting basis	Density (lbs/cu ft)	Moisture (%)	N (mg/kg)	P (mg/kg)	K (mg/kg)	Salt (mg/kg)	TFS (%)
04/22/2023	429.00 *ton*	Dry-weight		68.2	16,100.00	1,610.00	8,940.00		4.83

	Yield (tons/acre)	Total N (lbs/acre)	Total P (lbs/acre)	Total K (lbs/acre)	Salt (lbs/acre)
Anticipated harvest content	16.00	160.00	25.60	132.80	0.00
Total actual harvest content	13.00	133.11	13.31	73.92	399.34

05/15/2023: Corn, silage

Crop: Corn, silage Acres planted: 33 Plant date: 05/15/2023

Harvest date	Yield	Reporting basis	Density (lbs/cu ft)	Moisture (%)	N (mg/kg)	P (mg/kg)	K (mg/kg)	Salt (mg/kg)	TFS (%)
09/16/2023	1,056.00 *ton*	Dry-weight		68.2	16,100.00	1,610.00	8,940.00		4.83

	Yield (tons/acre)	Total N (lbs/acre)	Total P (lbs/acre)	Total K (lbs/acre)	Salt (lbs/acre)
Anticipated harvest content	30.00	240.00	45.00	249.00	0.00
Total actual harvest content	32.00	327.67	32.77	181.95	983.00

4

Field name: 4

11/01/2022: Oats, silage-soft dough

Crop: Oats, silage-soft dough Acres planted: 36 Plant date: 11/01/2022

Harvest date	Yield	Reporting basis	Density (lbs/cu ft)	Moisture (%)	N (mg/kg)	P (mg/kg)	K (mg/kg)	Salt (mg/kg)	TFS (%)
04/23/2023	468.00 *ton*	Dry-weight		57.6	7,210.00	1,230.00	12,200.00		12.40

	Yield (tons/acre)	Total N (lbs/acre)	Total P (lbs/acre)	Total K (lbs/acre)	Salt (lbs/acre)
Anticipated harvest content	16.00	160.00	25.60	132.80	0.00
Total actual harvest content	13.00	79.48	13.56	134.49	1,366.98

05/15/2023: Corn, silage

Crop: Corn, silage Acres planted: 36 Plant date: 05/15/2023

Harvest date	Yield	Reporting basis	Density (lbs/cu ft)	Moisture (%)	N (mg/kg)	P (mg/kg)	K (mg/kg)	Salt (mg/kg)	TFS (%)
09/17/2023	1,152.00 *ton*	Dry-weight		69.5	21,500.00	2,410.00	12,400.00		7.84

	Yield (tons/acre)	Total N (lbs/acre)	Total P (lbs/acre)	Total K (lbs/acre)	Salt (lbs/acre)
Anticipated harvest content	30.00	240.00	45.00	198.00	0.00
Total actual harvest content	32.00	419.68	47.04	242.05	1,530.37

5A

Field name: 5A

11/01/2022: Oats, silage-soft dough

Crop: Oats, silage-soft dough Acres planted: 14 Plant date: 11/01/2022

Harvest date	Yield	Reporting basis	Density (lbs/cu ft)	Moisture (%)	N (mg/kg)	P (mg/kg)	K (mg/kg)	Salt (mg/kg)	TFS (%)
04/24/2023	196.00 *ton*	Dry-weight		81.7	6,890.00	625.00	5,990.00		14.30

	Yield (tons/acre)	Total N (lbs/acre)	Total P (lbs/acre)	Total K (lbs/acre)	Salt (lbs/acre)
Anticipated harvest content	16.00	160.00	25.60	132.80	0.00
Total actual harvest content	14.00	35.30	3.20	30.69	732.73

5A

05/15/2023: Corn, silage

Crop: Corn, silage Acres planted: 14 Plant date: 05/15/2023

Harvest date	Yield	Reporting basis	Density (lbs/cu ft)	Moisture (%)	N (mg/kg)	P (mg/kg)	K (mg/kg)	Salt (mg/kg)	TFS (%)
09/18/2023	490.00 *ton*	Dry-weight		68.9	12,400.00	1,640.00	9,920.00		5.28

	Yield (tons/acre)	Total N (lbs/acre)	Total P (lbs/acre)	Total K (lbs/acre)	Salt (lbs/acre)
Anticipated harvest content	30.00	240.00	45.00	198.00	0.00
Total actual harvest content	35.00	269.95	35.70	215.96	1,149.46

5B

Field name: 5B

11/01/2022: Oats, silage-soft dough

Crop: Oats, silage-soft dough Acres planted: 12 Plant date: 11/01/2022

Harvest date	Yield	Reporting basis	Density (lbs/cu ft)	Moisture (%)	N (mg/kg)	P (mg/kg)	K (mg/kg)	Salt (mg/kg)	TFS (%)
04/24/2023	168.00 *ton*	Dry-weight		81.7	6,890.00	625.00	5,990.00		14.30

	Yield (tons/acre)	Total N (lbs/acre)	Total P (lbs/acre)	Total K (lbs/acre)	Salt (lbs/acre)
Anticipated harvest content	16.00	160.00	25.60	132.80	0.00
Total actual harvest content	14.00	35.30	3.20	30.69	732.73

05/15/2023: Corn, silage

Crop: Corn, silage Acres planted: 12 Plant date: 05/15/2023

Harvest date	Yield	Reporting basis	Density (lbs/cu ft)	Moisture (%)	N (mg/kg)	P (mg/kg)	K (mg/kg)	Salt (mg/kg)	TFS (%)
09/18/2023	420.00 *ton*	Dry-weight		68.9	12,400.00	1,640.00	9,920.00		5.28

	Yield (tons/acre)	Total N (lbs/acre)	Total P (lbs/acre)	Total K (lbs/acre)	Salt (lbs/acre)
Anticipated harvest content	30.00	240.00	45.00	198.00	0.00
Total actual harvest content	35.00	269.95	35.70	215.96	1,149.46

6A

Field name: 6A

11/01/2022: Oats, silage-soft dough

Crop: Oats, silage-soft dough Acres planted: 58 Plant date: 11/01/2022

Harvest date	Yield	Reporting basis	Density (lbs/cu ft)	Moisture (%)	N (mg/kg)	P (mg/kg)	K (mg/kg)	Salt (mg/kg)	TFS (%)
04/25/2023	841.00 *ton*	Dry-weight		75.8	5,160.00	582.00	5,380.00		11.80

	Yield (tons/acre)	Total N (lbs/acre)	Total P (lbs/acre)	Total K (lbs/acre)	Salt (lbs/acre)
Anticipated harvest content	16.00	160.00	25.60	132.80	0.00
Total actual harvest content	14.50	36.21	4.08	37.76	828.12

05/15/2023: Corn, silage

Crop: Corn, silage Acres planted: 58 Plant date: 05/15/2023

Harvest date	Yield	Reporting basis	Density (lbs/cu ft)	Moisture (%)	N (mg/kg)	P (mg/kg)	K (mg/kg)	Salt (mg/kg)	TFS (%)
09/19/2023	1,972.00 *ton*	Dry-weight		65.5	14,800.00	2,120.00	8,970.00		4.97

	Yield (tons/acre)	Total N (lbs/acre)	Total P (lbs/acre)	Total K (lbs/acre)	Salt (lbs/acre)
Anticipated harvest content	30.00	240.00	45.00	198.00	0.00
Total actual harvest content	34.00	347.21	49.74	210.44	1,165.96

6B

Field name: 6B

11/01/2022: Oats, silage-soft dough

Crop: Oats, silage-soft dough Acres planted: 7 Plant date: 11/01/2022

Harvest date	Yield	Reporting basis	Density (lbs/cu ft)	Moisture (%)	N (mg/kg)	P (mg/kg)	K (mg/kg)	Salt (mg/kg)	TFS (%)
04/25/2023	101.50 *ton*	Dry-weight		75.8	5,160.00	582.00	5,380.00		11.80

	Yield (tons/acre)	Total N (lbs/acre)	Total P (lbs/acre)	Total K (lbs/acre)	Salt (lbs/acre)
Anticipated harvest content	16.00	160.00	25.60	132.80	0.00
Total actual harvest content	14.50	36.21	4.08	37.76	828.12

6B

05/15/2023: Corn, silage

Crop: Corn, silage Acres planted: 7 Plant date: 05/15/2023

Harvest date	Yield	Reporting basis	Density (lbs/cu ft)	Moisture (%)	N (mg/kg)	P (mg/kg)	K (mg/kg)	Salt (mg/kg)	TFS (%)
09/19/2023	238.00 *ton*	Dry-weight		65.5	14,800.00	2,120.00	8,970.00		4.97

	Yield (tons/acre)	Total N (lbs/acre)	Total P (lbs/acre)	Total K (lbs/acre)	Salt (lbs/acre)
Anticipated harvest content	30.00	240.00	45.00	198.00	0.00
Total actual harvest content	34.00	347.21	49.74	210.44	1,165.96

7

Field name: 7

11/01/2022: Oats, silage-soft dough

Crop: Oats, silage-soft dough Acres planted: 31 Plant date: 11/01/2022

Harvest date	Yield	Reporting basis	Density (lbs/cu ft)	Moisture (%)	N (mg/kg)	P (mg/kg)	K (mg/kg)	Salt (mg/kg)	TFS (%)
04/26/2023	465.00 *ton*	Dry-weight		68.3	13,200.00	911.00	8,320.00		16.20

	Yield (tons/acre)	Total N (lbs/acre)	Total P (lbs/acre)	Total K (lbs/acre)	Salt (lbs/acre)
Anticipated harvest content	16.00	160.00	25.60	132.80	0.00
Total actual harvest content	15.00	125.53	8.66	79.12	1,540.62

05/15/2023: Corn, silage

Crop: Corn, silage Acres planted: 31 Plant date: 05/15/2023

Harvest date	Yield	Reporting basis	Density (lbs/cu ft)	Moisture (%)	N (mg/kg)	P (mg/kg)	K (mg/kg)	Salt (mg/kg)	TFS (%)
09/19/2023	1,085.00 *ton*	Dry-weight		65.5	14,800.00	2,120.00	8,970.00		4.97

	Yield (tons/acre)	Total N (lbs/acre)	Total P (lbs/acre)	Total K (lbs/acre)	Salt (lbs/acre)
Anticipated harvest content	30.00	240.00	45.00	198.00	0.00
Total actual harvest content	35.00	357.42	51.20	216.63	1,200.26

8

Field name: 8

11/01/2022: Oats, silage-soft dough

Crop: Oats, silage-soft dough Acres planted: 14 Plant date: 11/01/2022

Harvest date	Yield	Reporting basis	Density (lbs/cu ft)	Moisture (%)	N (mg/kg)	P (mg/kg)	K (mg/kg)	Salt (mg/kg)	TFS (%)
04/26/2023	203.00 *ton*	Dry-weight		75.7	8,620.00	794.00	8,110.00		15.40

	Yield (tons/acre)	Total N (lbs/acre)	Total P (lbs/acre)	Total K (lbs/acre)	Salt (lbs/acre)
Anticipated harvest content	16.00	160.00	25.60	132.80	0.00
Total actual harvest content	14.50	60.75	5.60	57.15	1,085.24

05/15/2023: Corn, silage

Crop: Corn, silage Acres planted: 14 Plant date: 05/15/2023

Harvest date	Yield	Reporting basis	Density (lbs/cu ft)	Moisture (%)	N (mg/kg)	P (mg/kg)	K (mg/kg)	Salt (mg/kg)	TFS (%)
09/20/2023	490.00 *ton*	Dry-weight		70.4	13,100.00	2,620.00	12,600.00		6.46

	Yield (tons/acre)	Total N (lbs/acre)	Total P (lbs/acre)	Total K (lbs/acre)	Salt (lbs/acre)
Anticipated harvest content	30.00	240.00	45.00	198.00	0.00
Total actual harvest content	35.00	271.43	54.29	261.07	1,338.51

9

Field name: 9

11/01/2022: Oats, silage-soft dough

Crop: Oats, silage-soft dough Acres planted: 18 Plant date: 11/01/2022

Harvest date	Yield	Reporting basis	Density (lbs/cu ft)	Moisture (%)	N (mg/kg)	P (mg/kg)	K (mg/kg)	Salt (mg/kg)	TFS (%)
04/26/2023	261.00 *ton*	Dry-weight		74.6	4,180.00	709.00	7,050.00		12.30

	Yield (tons/acre)	Total N (lbs/acre)	Total P (lbs/acre)	Total K (lbs/acre)	Salt (lbs/acre)
Anticipated harvest content	16.00	160.00	25.60	132.80	0.00
Total actual harvest content	14.50	30.79	5.22	51.93	906.02

9

05/15/2023: Corn, silage

Crop: Corn, silage Acres planted: 18 Plant date: 05/15/2023

Harvest date	Yield	Reporting basis	Density (lbs/cu ft)	Moisture (%)	N (mg/kg)	P (mg/kg)	K (mg/kg)	Salt (mg/kg)	TFS (%)
09/21/2023	612.00 *ton*	Dry-weight		0.1	0.00	0.00	0.00		0.00

	Yield (tons/acre)	Total N (lbs/acre)	Total P (lbs/acre)	Total K (lbs/acre)	Salt (lbs/acre)
Anticipated harvest content	30.00	240.00	45.00	198.00	0.00
Total actual harvest content	34.00	0.00	0.00	0.00	0.00

NUTRIENT BUDGET

A. LAND APPLICATIONS

1 - 05/15/2023: Corn, silage

Field name: 1

Crop: Corn, silage

Plant date: 05/15/2023

Application date	Application method	Precipitation 24 hours prior	Precipitation during application	Precipitation 24 hours following
04/22/2023	Surface (irrigation)	No precipitation	No precipitation	No precipitation

Source description	Material type	N (lbs/acre)	P (lbs/acre)	K (lbs/acre)	Salt (lbs/acre)	Amount
Lagoon Water	Process wastewater	38.06	12.27	53.02	53.02	1,343,925.00 *gal*
Canal Water	Surface water	0.58	0.00	0.00	96.29	4,479,750.00 *gal*
Application event totals		38.64	12.27	53.02	149.31	

Application date	Application method	Precipitation 24 hours prior	Precipitation during application	Precipitation 24 hours following
06/10/2023	Surface (irrigation)	No precipitation	No precipitation	No precipitation

Source description	Material type	N (lbs/acre)	P (lbs/acre)	K (lbs/acre)	Salt (lbs/acre)	Amount
Lagoon Water	Process wastewater	38.06	12.27	53.02	53.02	1,343,925.00 *gal*
Canal Water	Surface water	0.58	0.00	0.00	96.29	4,479,750.00 *gal*
Application event totals		38.64	12.27	53.02	149.31	

Application date	Application method	Precipitation 24 hours prior	Precipitation during application	Precipitation 24 hours following
06/23/2023	Surface (irrigation)	No precipitation	No precipitation	No precipitation

Source description	Material type	N (lbs/acre)	P (lbs/acre)	K (lbs/acre)	Salt (lbs/acre)	Amount
Lagoon Water	Process wastewater	20.30	6.54	28.28	28.28	716,760.00 *gal*
Canal Water	Surface water	0.40	0.00	0.00	67.40	3,135,825.00 *gal*
Application event totals		20.70	6.54	28.28	95.68	

Application date	Application method	Precipitation 24 hours prior	Precipitation during application	Precipitation 24 hours following
07/05/2023	Surface (irrigation)	No precipitation	No precipitation	No precipitation

Source description	Material type	N (lbs/acre)	P (lbs/acre)	K (lbs/acre)	Salt (lbs/acre)	Amount
Lagoon Water	Process wastewater	20.30	6.54	28.28	28.28	716,760.00 *gal*
Canal Water	Surface water	0.40	0.00	0.00	67.40	3,135,825.00 *gal*
Application event totals		20.70	6.54	28.28	95.68	

1 - 05/15/2023: Corn, silage

Application date	Application method	Precipitation 24 hours prior	Precipitation during application	Precipitation 24 hours following
07/17/2023	Surface (irrigation)	No precipitation	No precipitation	No precipitation

Source description	Material type	N (lbs/acre)	P (lbs/acre)	K (lbs/acre)	Salt (lbs/acre)	Amount
Lagoon Water	Process wastewater	20.30	6.54	28.28	28.28	716,760.00 *gal*
Canal Water	Surface water	0.40	0.00	0.00	67.40	3,135,825.00 *gal*
Application event totals		20.70	6.54	28.28	95.68	

Application date	Application method	Precipitation 24 hours prior	Precipitation during application	Precipitation 24 hours following
07/29/2023	Surface (irrigation)	No precipitation	No precipitation	No precipitation

Source description	Material type	N (lbs/acre)	P (lbs/acre)	K (lbs/acre)	Salt (lbs/acre)	Amount
Lagoon Water	Process wastewater	20.30	6.54	28.28	28.28	716,760.00 *gal*
Canal Water	Surface water	0.40	0.00	0.00	67.40	3,135,825.00 *gal*
Application event totals		20.70	6.54	28.28	95.68	

Application date	Application method	Precipitation 24 hours prior	Precipitation during application	Precipitation 24 hours following
08/10/2023	Surface (irrigation)	No precipitation	No precipitation	No precipitation

Source description	Material type	N (lbs/acre)	P (lbs/acre)	K (lbs/acre)	Salt (lbs/acre)	Amount
Lagoon Water	Process wastewater	20.30	6.54	28.28	28.28	716,760.00 *gal*
Canal Water	Surface water	0.40	0.00	0.00	67.40	3,135,825.00 *gal*
Application event totals		20.70	6.54	28.28	95.68	

Application date	Application method	Precipitation 24 hours prior	Precipitation during application	Precipitation 24 hours following
08/22/2023	Surface (irrigation)	No precipitation	No precipitation	No precipitation

Source description	Material type	N (lbs/acre)	P (lbs/acre)	K (lbs/acre)	Salt (lbs/acre)	Amount
Lagoon Water	Process wastewater	50.03	11.82	75.40	453.13	716,760.00 *gal*
Canal Water	Surface water	0.40	0.00	0.00	67.40	3,135,825.00 *gal*
Application event totals		50.43	11.82	75.40	520.54	

2A - 05/15/2023: Corn, silage

Field name: 2A

Crop: Corn, silage

Plant date: 05/15/2023

Application date	Application method	Precipitation 24 hours prior	Precipitation during application	Precipitation 24 hours following

2A - 05/15/2023: Corn, silage

Application date	Application method	Precipitation 24 hours prior	Precipitation during application	Precipitation 24 hours following
04/23/2023	Surface (irrigation)	No precipitation	No precipitation	No precipitation

Source description	Material type	N (lbs/acre)	P (lbs/acre)	K (lbs/acre)	Salt (lbs/acre)	Amount
Lagoon Water	Process wastewater	38.06	12.27	53.02	53.02	1,588,275.00 *gal*
Canal Water	Surface water	0.58	0.00	0.00	96.29	5,294,250.00 *gal*
Application event totals		38.64	12.27	53.02	149.31	

06/11/2023	Surface (irrigation)	No precipitation	No precipitation	No precipitation

Source description	Material type	N (lbs/acre)	P (lbs/acre)	K (lbs/acre)	Salt (lbs/acre)	Amount
Lagoon Water	Process wastewater	38.06	12.27	53.02	53.02	1,588,275.00 *gal*
Canal Water	Surface water	0.58	0.00	0.00	96.29	5,294,250.00 *gal*
Application event totals		38.64	12.27	53.02	149.31	

06/24/2023	Surface (irrigation)	No precipitation	No precipitation	No precipitation

Source description	Material type	N (lbs/acre)	P (lbs/acre)	K (lbs/acre)	Salt (lbs/acre)	Amount
Lagoon Water	Process wastewater	20.30	6.54	28.28	28.28	847,080.00 *gal*
Canal Water	Surface water	0.40	0.00	0.00	67.40	3,705,975.00 *gal*
Application event totals		20.70	6.54	28.28	95.68	

07/06/2023	Surface (irrigation)	No precipitation	No precipitation	No precipitation

Source description	Material type	N (lbs/acre)	P (lbs/acre)	K (lbs/acre)	Salt (lbs/acre)	Amount
Lagoon Water	Process wastewater	20.30	6.54	28.28	28.28	847,080.00 *gal*
Canal Water	Surface water	0.40	0.00	0.00	67.40	3,705,975.00 *gal*
Application event totals		20.70	6.54	28.28	95.68	

07/18/2023	Surface (irrigation)	No precipitation	No precipitation	No precipitation

Source description	Material type	N (lbs/acre)	P (lbs/acre)	K (lbs/acre)	Salt (lbs/acre)	Amount
Lagoon Water	Process wastewater	20.30	6.54	28.28	28.28	847,080.00 *gal*
Canal Water	Surface water	0.40	0.00	0.00	67.40	3,705,975.00 *gal*
Application event totals		20.70	6.54	28.28	95.68	

2A - 05/15/2023: Corn, silage

Application date	Application method	Precipitation 24 hours prior	Precipitation during application	Precipitation 24 hours following
07/30/2023	Surface (irrigation)	No precipitation	No precipitation	No precipitation

Source description	Material type	N (lbs/acre)	P (lbs/acre)	K (lbs/acre)	Salt (lbs/acre)	Amount
Lagoon Water	Process wastewater	20.30	6.54	28.28	28.28	847,080.00 *gal*
Canal Water	Surface water	0.40	0.00	0.00	67.40	3,705,975.00 *gal*
Application event totals		20.70	6.54	28.28	95.68	

Application date	Application method	Precipitation 24 hours prior	Precipitation during application	Precipitation 24 hours following
08/11/2023	Surface (irrigation)	No precipitation	No precipitation	No precipitation

Source description	Material type	N (lbs/acre)	P (lbs/acre)	K (lbs/acre)	Salt (lbs/acre)	Amount
Lagoon Water	Process wastewater	20.30	6.54	28.28	28.28	847,080.00 *gal*
Canal Water	Surface water	0.40	0.00	0.00	67.40	3,705,975.00 *gal*
Application event totals		20.70	6.54	28.28	95.68	

Application date	Application method	Precipitation 24 hours prior	Precipitation during application	Precipitation 24 hours following
08/23/2023	Surface (irrigation)	No precipitation	No precipitation	No precipitation

Source description	Material type	N (lbs/acre)	P (lbs/acre)	K (lbs/acre)	Salt (lbs/acre)	Amount
Lagoon Water	Process wastewater	50.03	11.82	75.40	453.13	847,080.00 *gal*
Canal Water	Surface water	0.40	0.00	0.00	67.40	3,705,975.00 *gal*
Application event totals		50.43	11.82	75.40	520.54	

2B - 05/15/2023: Corn, silage

Field name: 2B

Crop: Corn, silage

Plant date: 05/15/2023

Application date	Application method	Precipitation 24 hours prior	Precipitation during application	Precipitation 24 hours following
04/23/2023	Surface (irrigation)	No precipitation	No precipitation	No precipitation

Source description	Material type	N (lbs/acre)	P (lbs/acre)	K (lbs/acre)	Salt (lbs/acre)	Amount
Lagoon Water	Process wastewater	38.06	12.27	53.02	53.02	203,625.00 *gal*
Canal Water	Surface water	0.58	0.00	0.00	96.29	678,750.00 *gal*
Application event totals		38.64	12.27	53.02	149.31	

2B - 05/15/2023: Corn, silage

Application date	Application method	Precipitation 24 hours prior	Precipitation during application	Precipitation 24 hours following
06/11/2023	Surface (irrigation)	No precipitation	No precipitation	No precipitation

Source description	Material type	N (lbs/acre)	P (lbs/acre)	K (lbs/acre)	Salt (lbs/acre)	Amount
Lagoon Water	Process wastewater	38.06	12.27	53.02	53.02	203,625.00 *gal*
Canal Water	Surface water	0.58	0.00	0.00	96.29	678,750.00 *gal*
Application event totals		38.64	12.27	53.02	149.31	

Application date	Application method	Precipitation 24 hours prior	Precipitation during application	Precipitation 24 hours following
06/24/2023	Surface (irrigation)	No precipitation	No precipitation	No precipitation

Source description	Material type	N (lbs/acre)	P (lbs/acre)	K (lbs/acre)	Salt (lbs/acre)	Amount
Lagoon Water	Process wastewater	20.19	6.51	28.12	28.12	108,000.00 *gal*
Canal Water	Surface water	0.40	0.00	0.00	67.40	475,125.00 *gal*
Application event totals		20.59	6.51	28.12	95.52	

Application date	Application method	Precipitation 24 hours prior	Precipitation during application	Precipitation 24 hours following
07/06/2023	Surface (irrigation)	No precipitation	No precipitation	No precipitation

Source description	Material type	N (lbs/acre)	P (lbs/acre)	K (lbs/acre)	Salt (lbs/acre)	Amount
Lagoon Water	Process wastewater	20.30	6.54	28.28	28.28	108,600.00 *gal*
Canal Water	Surface water	0.40	0.00	0.00	67.40	475,125.00 *gal*
Application event totals		20.70	6.54	28.28	95.68	

Application date	Application method	Precipitation 24 hours prior	Precipitation during application	Precipitation 24 hours following
07/18/2023	Surface (irrigation)	No precipitation	No precipitation	No precipitation

Source description	Material type	N (lbs/acre)	P (lbs/acre)	K (lbs/acre)	Salt (lbs/acre)	Amount
Lagoon Water	Process wastewater	20.30	6.54	28.28	28.28	108,600.00 *gal*
Canal Water	Surface water	0.40	0.00	0.00	67.40	475,125.00 *gal*
Application event totals		20.70	6.54	28.28	95.68	

Application date	Application method	Precipitation 24 hours prior	Precipitation during application	Precipitation 24 hours following
07/30/2023	Surface (irrigation)	No precipitation	No precipitation	No precipitation

Source description	Material type	N (lbs/acre)	P (lbs/acre)	K (lbs/acre)	Salt (lbs/acre)	Amount
Lagoon Water	Process wastewater	20.30	6.54	28.28	28.28	108,600.00 *gal*
Canal Water	Surface water	0.40	0.00	0.00	67.40	475,125.00 *gal*
Application event totals		20.70	6.54	28.28	95.68	

2B - 05/15/2023: Corn, silage

Application date	Application method	Precipitation 24 hours prior	Precipitation during application	Precipitation 24 hours following
08/11/2023	Surface (irrigation)	No precipitation	No precipitation	No precipitation

Source description	Material type	N (lbs/acre)	P (lbs/acre)	K (lbs/acre)	Salt (lbs/acre)	Amount
Lagoon Water	Process wastewater	20.30	6.54	28.28	28.28	108,600.00 *gal*
Canal Water	Surface water	0.40	0.00	0.00	67.40	475,125.00 *gal*
Application event totals		20.70	6.54	28.28	95.68	

Application date	Application method	Precipitation 24 hours prior	Precipitation during application	Precipitation 24 hours following
08/24/2023	Surface (irrigation)	No precipitation	No precipitation	No precipitation

Source description	Material type	N (lbs/acre)	P (lbs/acre)	K (lbs/acre)	Salt (lbs/acre)	Amount
Lagoon Water	Process wastewater	20.30	6.54	28.28	28.28	108,600.00 *gal*
Canal Water	Surface water	0.40	0.00	0.00	67.40	475,125.00 *gal*
Application event totals		20.70	6.54	28.28	95.68	

3 - 11/01/2022: Oats, silage-soft dough

Field name: 3

Crop: Oats, silage-soft dough

Plant date: 11/01/2022

Application date	Application method	Precipitation 24 hours prior	Precipitation during application	Precipitation 24 hours following
02/27/2023	Surface (irrigation)	No precipitation	No precipitation	No precipitation

Source description	Material type	N (lbs/acre)	P (lbs/acre)	K (lbs/acre)	Salt (lbs/acre)	Amount
Lagoon Water	Process wastewater	111.06	18.75	119.26	673.67	720,000.00 *gal*
Application event totals		111.06	18.75	119.26	673.67	

Application date	Application method	Precipitation 24 hours prior	Precipitation during application	Precipitation 24 hours following
03/06/2023	Surface (irrigation)	No precipitation	No precipitation	No precipitation

Source description	Material type	N (lbs/acre)	P (lbs/acre)	K (lbs/acre)	Salt (lbs/acre)	Amount
Lagoon Water	Process wastewater	66.01	14.14	121.51	470.35	620,000.00 *gal*
Application event totals		66.01	14.14	121.51	470.35	

3 - 05/15/2023: Corn, silage

3 - 05/15/2023: Corn, silage

Field name: 3

Crop: Corn, silage

Plant date: 05/15/2023

Application date	Application method	Precipitation 24 hours prior	Precipitation during application	Precipitation 24 hours following
04/24/2023	Surface (irrigation)	No precipitation	No precipitation	No precipitation

Source description	Material type	N (lbs/acre)	P (lbs/acre)	K (lbs/acre)	Salt (lbs/acre)	Amount
Lagoon Water	Process wastewater	38.06	12.27	53.02	53.02	1,343,925.00 *gal*
Canal Water	Surface water	0.58	0.00	0.00	96.29	4,479,750.00 *gal*
Application event totals		38.64	12.27	53.02	149.31	

06/12/2023	Surface (irrigation)	No precipitation	No precipitation	No precipitation

Source description	Material type	N (lbs/acre)	P (lbs/acre)	K (lbs/acre)	Salt (lbs/acre)	Amount
Lagoon Water	Process wastewater	38.06	12.27	53.02	53.02	1,343,925.00 *gal*
Canal Water	Surface water	0.58	0.00	0.00	96.29	4,479,750.00 *gal*
Application event totals		38.64	12.27	53.02	149.31	

06/25/2023	Surface (irrigation)	No precipitation	No precipitation	No precipitation

Source description	Material type	N (lbs/acre)	P (lbs/acre)	K (lbs/acre)	Salt (lbs/acre)	Amount
Lagoon Water	Process wastewater	20.30	6.54	28.28	28.28	716,760.00 *gal*
Canal Water	Surface water	0.40	0.00	0.00	67.40	3,135,825.00 *gal*
Application event totals		20.70	6.54	28.28	95.68	

07/07/2023	Surface (irrigation)	No precipitation	No precipitation	No precipitation

Source description	Material type	N (lbs/acre)	P (lbs/acre)	K (lbs/acre)	Salt (lbs/acre)	Amount
Lagoon Water	Process wastewater	20.30	6.54	28.28	28.28	716,760.00 *gal*
Canal Water	Surface water	0.40	0.00	0.00	67.40	3,135,825.00 *gal*
Application event totals		20.70	6.54	28.28	95.68	

07/19/2023	Surface (irrigation)	No precipitation	No precipitation	No precipitation

Source description	Material type	N (lbs/acre)	P (lbs/acre)	K (lbs/acre)	Salt (lbs/acre)	Amount
Lagoon Water	Process wastewater	20.30	6.54	28.28	28.28	716,760.00 *gal*
Canal Water	Surface water	0.40	0.00	0.00	67.40	3,135,825.00 *gal*
Application event totals		20.70	6.54	28.28	95.68	

3 - 05/15/2023: Corn, silage

Application date	Application method	Precipitation 24 hours prior	Precipitation during application	Precipitation 24 hours following
07/31/2023	Surface (irrigation)	No precipitation	No precipitation	No precipitation

Source description	Material type	N (lbs/acre)	P (lbs/acre)	K (lbs/acre)	Salt (lbs/acre)	Amount
Lagoon Water	Process wastewater	20.30	6.54	28.28	28.28	716,760.00 *gal*
Canal Water	Surface water	0.40	0.00	0.00	67.40	3,135,825.00 *gal*
Application event totals		20.70	6.54	28.28	95.68	

08/12/2023	Surface (irrigation)	No precipitation	No precipitation	No precipitation

Source description	Material type	N (lbs/acre)	P (lbs/acre)	K (lbs/acre)	Salt (lbs/acre)	Amount
Lagoon Water	Process wastewater	20.30	6.54	28.28	28.28	716,760.00 *gal*
Canal Water	Surface water	0.40	0.00	0.00	67.40	3,135,825.00 *gal*
Application event totals		20.70	6.54	28.28	95.68	

08/25/2023	Surface (irrigation)	No precipitation	No precipitation	No precipitation

Source description	Material type	N (lbs/acre)	P (lbs/acre)	K (lbs/acre)	Salt (lbs/acre)	Amount
Lagoon Water	Process wastewater	20.30	6.54	28.28	28.28	716,760.00 *gal*
Canal Water	Surface water	0.40	0.00	0.00	67.40	3,135,825.00 *gal*
Application event totals		20.70	6.54	28.28	95.68	

4 - 11/01/2022: Oats, silage-soft dough

Field name: 4

Crop: Oats, silage-soft dough

Plant date: 11/01/2022

Application date	Application method	Precipitation 24 hours prior	Precipitation during application	Precipitation 24 hours following
03/13/2023	Surface (irrigation)	No precipitation	No precipitation	No precipitation

Source description	Material type	N (lbs/acre)	P (lbs/acre)	K (lbs/acre)	Salt (lbs/acre)	Amount
Lagoon Water	Process wastewater	50.75	10.87	93.42	361.62	520,000.00 *gal*
Application event totals		50.75	10.87	93.42	361.62	

4 - 11/01/2022: Oats, silage-soft dough

Application date	Application method	Precipitation 24 hours prior	Precipitation during application	Precipitation 24 hours following
03/20/2023	Surface (irrigation)	No precipitation	No precipitation	No precipitation

Source description	Material type	N (lbs/acre)	P (lbs/acre)	K (lbs/acre)	Salt (lbs/acre)	Amount
Lagoon Water	Process wastewater	50.75	10.87	93.42	361.62	520,000.00 *gal*
Application event totals		50.75	10.87	93.42	361.62	

4 - 05/15/2023: Corn, silage

Field name: 4

Crop: Corn, silage

Plant date: 05/15/2023

Application date	Application method	Precipitation 24 hours prior	Precipitation during application	Precipitation 24 hours following
04/25/2023	Surface (irrigation)	No precipitation	No precipitation	No precipitation

Source description	Material type	N (lbs/acre)	P (lbs/acre)	K (lbs/acre)	Salt (lbs/acre)	Amount
Lagoon Water	Process wastewater	38.06	12.27	53.02	53.02	1,466,100.00 *gal*
Canal Water	Surface water	0.58	0.00	0.00	96.29	4,887,000.00 *gal*
Application event totals		38.64	12.27	53.02	149.31	

Application date	Application method	Precipitation 24 hours prior	Precipitation during application	Precipitation 24 hours following
06/13/2023	Surface (irrigation)	No precipitation	No precipitation	No precipitation

Source description	Material type	N (lbs/acre)	P (lbs/acre)	K (lbs/acre)	Salt (lbs/acre)	Amount
Lagoon Water	Process wastewater	38.06	12.27	53.02	53.02	1,466,100.00 *gal*
Canal Water	Surface water	0.58	0.00	0.00	96.29	4,887,000.00 *gal*
Application event totals		38.64	12.27	53.02	149.31	

Application date	Application method	Precipitation 24 hours prior	Precipitation during application	Precipitation 24 hours following
06/26/2023	Surface (irrigation)	No precipitation	No precipitation	No precipitation

Source description	Material type	N (lbs/acre)	P (lbs/acre)	K (lbs/acre)	Salt (lbs/acre)	Amount
Lagoon Water	Process wastewater	20.30	6.54	28.28	28.28	781,920.00 *gal*
Canal Water	Surface water	0.40	0.00	0.00	67.40	3,420,900.00 *gal*
Application event totals		20.70	6.54	28.28	95.68	

4 - 05/15/2023: Corn, silage

Application date	Application method	Precipitation 24 hours prior	Precipitation during application	Precipitation 24 hours following
07/05/2023	Surface (irrigation)	No precipitation	No precipitation	No precipitation

Source description	Material type	N (lbs/acre)	P (lbs/acre)	K (lbs/acre)	Salt (lbs/acre)	Amount
Lagoon Water	Process wastewater	20.30	6.54	28.28	28.28	781,920.00 *gal*
Canal Water	Surface water	0.40	0.00	0.00	67.40	3,420,900.00 *gal*
Application event totals		20.70	6.54	28.28	95.68	

07/20/2023	Surface (irrigation)	No precipitation	No precipitation	No precipitation

Source description	Material type	N (lbs/acre)	P (lbs/acre)	K (lbs/acre)	Salt (lbs/acre)	Amount
Lagoon Water	Process wastewater	20.30	6.54	28.28	28.28	781,920.00 *gal*
Canal Water	Surface water	0.40	0.00	0.00	67.40	3,420,900.00 *gal*
Application event totals		20.70	6.54	28.28	95.68	

08/01/2023	Surface (irrigation)	No precipitation	No precipitation	No precipitation

Source description	Material type	N (lbs/acre)	P (lbs/acre)	K (lbs/acre)	Salt (lbs/acre)	Amount
Lagoon Water	Process wastewater	20.30	6.54	28.28	174.00	781,920.00 *gal*
Canal Water	Surface water	0.40	0.00	0.00	67.40	3,420,900.00 *gal*
Application event totals		20.70	6.54	28.28	241.41	

08/13/2023	Surface (irrigation)	No precipitation	No precipitation	No precipitation

Source description	Material type	N (lbs/acre)	P (lbs/acre)	K (lbs/acre)	Salt (lbs/acre)	Amount
Lagoon Water	Process wastewater	20.30	6.54	28.28	174.00	781,920.00 *gal*
Canal Water	Surface water	0.40	0.00	0.00	67.40	3,420,900.00 *gal*
Application event totals		20.70	6.54	28.28	241.41	

08/26/2023	Surface (irrigation)	No precipitation	No precipitation	No precipitation

Source description	Material type	N (lbs/acre)	P (lbs/acre)	K (lbs/acre)	Salt (lbs/acre)	Amount
Lagoon Water	Process wastewater	20.30	6.54	28.28	174.00	781,920.00 *gal*
Canal Water	Surface water	0.40	0.00	0.00	67.40	3,420,900.00 *gal*
Application event totals		20.70	6.54	28.28	241.41	

5A - 11/01/2022: Oats, silage-soft dough

5A - 11/01/2022: Oats, silage-soft dough

Field name: 5A

Crop: Oats, silage-soft dough

Plant date: 11/01/2022

Application date	Application method	Precipitation 24 hours prior	Precipitation during application	Precipitation 24 hours following
02/13/2023	Surface (irrigation)	No precipitation	No precipitation	No precipitation

Source description	Material type	N (lbs/acre)	P (lbs/acre)	K (lbs/acre)	Salt (lbs/acre)	Amount
Lagoon Water	Process wastewater	20.08	4.30	36.96	143.06	80,000.00 *gal*
Application event totals		20.08	4.30	36.96	143.06	

Application date	Application method	Precipitation 24 hours prior	Precipitation during application	Precipitation 24 hours following
02/20/2023	Surface (irrigation)	No precipitation	No precipitation	No precipitation

Source description	Material type	N (lbs/acre)	P (lbs/acre)	K (lbs/acre)	Salt (lbs/acre)	Amount
Lagoon Water	Process wastewater	20.08	4.30	36.96	143.06	80,000.00 *gal*
Application event totals		20.08	4.30	36.96	143.06	

5A - 05/15/2023: Corn, silage

Field name: 5A

Crop: Corn, silage

Plant date: 05/15/2023

Application date	Application method	Precipitation 24 hours prior	Precipitation during application	Precipitation 24 hours following
04/26/2023	Surface (irrigation)	No precipitation	No precipitation	No precipitation

Source description	Material type	N (lbs/acre)	P (lbs/acre)	K (lbs/acre)	Salt (lbs/acre)	Amount
Lagoon Water	Process wastewater	38.06	12.27	53.02	326.26	570,150.00 *gal*
Canal Water	Surface water	0.58	0.00	0.00	96.29	1,900,500.00 *gal*
Application event totals		38.64	12.27	53.02	422.55	

Application date	Application method	Precipitation 24 hours prior	Precipitation during application	Precipitation 24 hours following
06/14/2023	Surface (irrigation)	No precipitation	No precipitation	No precipitation

Source description	Material type	N (lbs/acre)	P (lbs/acre)	K (lbs/acre)	Salt (lbs/acre)	Amount
Lagoon Water	Process wastewater	38.06	12.27	53.02	326.26	570,150.00 *gal*
Canal Water	Surface water	0.58	0.00	0.00	96.29	1,900,500.00 *gal*
Application event totals		38.64	12.27	53.02	422.55	

5A - 05/15/2023: Corn, silage

Application date	Application method	Precipitation 24 hours prior	Precipitation during application	Precipitation 24 hours following
06/27/2023	Surface (irrigation)	No precipitation	No precipitation	No precipitation

Source description	Material type	N (lbs/acre)	P (lbs/acre)	K (lbs/acre)	Salt (lbs/acre)	Amount
Lagoon Water	Process wastewater	20.30	6.54	28.28	174.00	304,080.00 *gal*
Canal Water	Surface water	0.40	0.00	0.00	67.40	1,330,350.00 *gal*
Application event totals		20.70	6.54	28.28	241.41	

Application date	Application method	Precipitation 24 hours prior	Precipitation during application	Precipitation 24 hours following
07/09/2023	Surface (irrigation)	No precipitation	No precipitation	No precipitation

Source description	Material type	N (lbs/acre)	P (lbs/acre)	K (lbs/acre)	Salt (lbs/acre)	Amount
Lagoon Water	Process wastewater	20.30	6.54	28.28	174.00	304,080.00 *gal*
Canal Water	Surface water	0.40	0.00	0.00	67.40	1,330,350.00 *gal*
Application event totals		20.70	6.54	28.28	241.41	

Application date	Application method	Precipitation 24 hours prior	Precipitation during application	Precipitation 24 hours following
07/21/2023	Surface (irrigation)	No precipitation	No precipitation	No precipitation

Source description	Material type	N (lbs/acre)	P (lbs/acre)	K (lbs/acre)	Salt (lbs/acre)	Amount
Lagoon Water	Process wastewater	20.30	6.54	28.28	174.00	304,080.00 *gal*
Canal Water	Surface water	0.40	0.00	0.00	67.40	1,330,350.00 *gal*
Application event totals		20.70	6.54	28.28	241.41	

Application date	Application method	Precipitation 24 hours prior	Precipitation during application	Precipitation 24 hours following
08/02/2023	Surface (irrigation)	No precipitation	No precipitation	No precipitation

Source description	Material type	N (lbs/acre)	P (lbs/acre)	K (lbs/acre)	Salt (lbs/acre)	Amount
Lagoon Water	Process wastewater	20.30	6.54	28.28	174.00	304,080.00 *gal*
Canal Water	Surface water	0.40	0.00	0.00	67.40	1,330,350.00 *gal*
Application event totals		20.70	6.54	28.28	241.41	

Application date	Application method	Precipitation 24 hours prior	Precipitation during application	Precipitation 24 hours following
08/14/2023	Surface (irrigation)	No precipitation	No precipitation	No precipitation

Source description	Material type	N (lbs/acre)	P (lbs/acre)	K (lbs/acre)	Salt (lbs/acre)	Amount
Lagoon Water	Process wastewater	20.30	6.54	28.28	174.00	304,080.00 *gal*
Canal Water	Surface water	0.40	0.00	0.00	67.40	1,330,350.00 *gal*
Application event totals		20.70	6.54	28.28	241.41	

5A - 05/15/2023: Corn, silage

Application date	Application method	Precipitation 24 hours prior	Precipitation during application	Precipitation 24 hours following
08/26/2023	Surface (irrigation)	No precipitation	No precipitation	No precipitation

Source description	Material type	N (lbs/acre)	P (lbs/acre)	K (lbs/acre)	Salt (lbs/acre)	Amount
Lagoon Water	Process wastewater	20.30	6.54	28.28	174.00	304,080.00 *gal*
Canal Water	Surface water	0.40	0.00	0.00	67.40	1,330,350.00 *gal*
Application event totals		20.70	6.54	28.28	241.41	

5B - 11/01/2022: Oats, silage-soft dough

Field name: 5B

Crop: Oats, silage-soft dough

Plant date: 11/01/2022

Application date	Application method	Precipitation 24 hours prior	Precipitation during application	Precipitation 24 hours following
02/13/2023	Surface (irrigation)	No precipitation	No precipitation	No precipitation

Source description	Material type	N (lbs/acre)	P (lbs/acre)	K (lbs/acre)	Salt (lbs/acre)	Amount
Lagoon Water	Process wastewater	21.21	3.58	22.77	128.65	50,000.00 *gal*
Application event totals		21.21	3.58	22.77	128.65	

Application date	Application method	Precipitation 24 hours prior	Precipitation during application	Precipitation 24 hours following
02/20/2023	Surface (irrigation)	No precipitation	No precipitation	No precipitation

Source description	Material type	N (lbs/acre)	P (lbs/acre)	K (lbs/acre)	Salt (lbs/acre)	Amount
Lagoon Water	Process wastewater	17.57	3.76	32.34	125.18	60,000.00 *gal*
Application event totals		17.57	3.76	32.34	125.18	

5B - 05/15/2023: Corn, silage

Field name: 5B

Crop: Corn, silage

Plant date: 05/15/2023

Application date	Application method	Precipitation 24 hours prior	Precipitation during application	Precipitation 24 hours following

5B - 05/15/2023: Corn, silage

Application date	Application method	Precipitation 24 hours prior	Precipitation during application	Precipitation 24 hours following
04/26/2023	Surface (irrigation)	No precipitation	No precipitation	No precipitation

Source description	Material type	N (lbs/acre)	P (lbs/acre)	K (lbs/acre)	Salt (lbs/acre)	Amount
Lagoon Water	Process wastewater	38.06	12.27	53.02	326.26	488,700.00 *gal*
Canal Water	Surface water	0.58	0.00	0.00	96.29	1,629,000.00 *gal*
Application event totals		38.64	12.27	53.02	422.55	

Application date	Application method	Precipitation 24 hours prior	Precipitation during application	Precipitation 24 hours following
06/14/2023	Surface (irrigation)	No precipitation	No precipitation	No precipitation

Source description	Material type	N (lbs/acre)	P (lbs/acre)	K (lbs/acre)	Salt (lbs/acre)	Amount
Lagoon Water	Process wastewater	38.06	12.27	53.02	326.26	488,700.00 *gal*
Canal Water	Surface water	0.58	0.00	0.00	96.29	1,629,000.00 *gal*
Application event totals		38.64	12.27	53.02	422.55	

Application date	Application method	Precipitation 24 hours prior	Precipitation during application	Precipitation 24 hours following
06/27/2023	Surface (irrigation)	No precipitation	No precipitation	No precipitation

Source description	Material type	N (lbs/acre)	P (lbs/acre)	K (lbs/acre)	Salt (lbs/acre)	Amount
Lagoon Water	Process wastewater	20.30	6.54	28.28	174.00	260,640.00 *gal*
Canal Water	Surface water	0.40	0.00	0.00	67.40	1,140,300.00 *gal*
Application event totals		20.70	6.54	28.28	241.41	

Application date	Application method	Precipitation 24 hours prior	Precipitation during application	Precipitation 24 hours following
07/09/2023	Surface (irrigation)	No precipitation	No precipitation	No precipitation

Source description	Material type	N (lbs/acre)	P (lbs/acre)	K (lbs/acre)	Salt (lbs/acre)	Amount
Lagoon Water	Process wastewater	20.30	6.54	28.28	174.00	260,640.00 *gal*
Canal Water	Surface water	0.40	0.00	0.00	67.40	1,140,300.00 *gal*
Application event totals		20.70	6.54	28.28	241.41	

Application date	Application method	Precipitation 24 hours prior	Precipitation during application	Precipitation 24 hours following
07/21/2023	Surface (irrigation)	No precipitation	No precipitation	No precipitation

Source description	Material type	N (lbs/acre)	P (lbs/acre)	K (lbs/acre)	Salt (lbs/acre)	Amount
Lagoon Water	Process wastewater	20.30	6.54	28.28	174.00	260,640.00 *gal*
Canal Water	Surface water	0.40	0.00	0.00	67.40	1,140,300.00 *gal*
Application event totals		20.70	6.54	28.28	241.41	

5B - 05/15/2023: Corn, silage

Application date	Application method	Precipitation 24 hours prior	Precipitation during application	Precipitation 24 hours following
08/02/2023	Surface (irrigation)	No precipitation	No precipitation	No precipitation

Source description	Material type	N (lbs/acre)	P (lbs/acre)	K (lbs/acre)	Salt (lbs/acre)	Amount
Lagoon Water	Process wastewater	20.30	6.54	28.28	174.00	260,640.00 *gal*
Canal Water	Surface water	0.40	0.00	0.00	67.40	1,140,300.00 *gal*
Application event totals		20.70	6.54	28.28	241.41	

Application date	Application method	Precipitation 24 hours prior	Precipitation during application	Precipitation 24 hours following
08/14/2023	Surface (irrigation)	No precipitation	No precipitation	No precipitation

Source description	Material type	N (lbs/acre)	P (lbs/acre)	K (lbs/acre)	Salt (lbs/acre)	Amount
Lagoon Water	Process wastewater	20.30	6.54	28.28	174.00	260,640.00 *gal*
Canal Water	Surface water	0.40	0.00	0.00	67.40	1,140,300.00 *gal*
Application event totals		20.70	6.54	28.28	241.41	

Application date	Application method	Precipitation 24 hours prior	Precipitation during application	Precipitation 24 hours following
08/26/2023	Surface (irrigation)	No precipitation	No precipitation	No precipitation

Source description	Material type	N (lbs/acre)	P (lbs/acre)	K (lbs/acre)	Salt (lbs/acre)	Amount
Lagoon Water	Process wastewater	20.30	6.54	28.28	174.00	260,640.00 *gal*
Canal Water	Surface water	0.40	0.00	0.00	67.40	1,140,300.00 *gal*
Application event totals		20.70	6.54	28.28	241.41	

6A - 11/01/2022: Oats, silage-soft dough

Field name: 6A

Crop: Oats, silage-soft dough

Plant date: 11/01/2022

Application date	Application method	Precipitation 24 hours prior	Precipitation during application	Precipitation 24 hours following
01/16/2023	Surface (irrigation)	No precipitation	No precipitation	No precipitation

Source description	Material type	N (lbs/acre)	P (lbs/acre)	K (lbs/acre)	Salt (lbs/acre)	Amount
Lagoon Water	Process wastewater	12.50	2.11	13.42	75.81	142,408.00 *gal*
Application event totals		12.50	2.11	13.42	75.81	

6A - 11/01/2022: Oats, silage-soft dough

Application date	Application method	Precipitation 24 hours prior	Precipitation during application	Precipitation 24 hours following
01/23/2023	Surface (irrigation)	No precipitation	No precipitation	No precipitation

Source description	Material type	N (lbs/acre)	P (lbs/acre)	K (lbs/acre)	Salt (lbs/acre)	Amount
Lagoon Water	Process wastewater	12.50	2.11	13.42	75.81	142,408.00 *gal*
Application event totals		12.50	2.11	13.42	75.81	

Application date	Application method	Precipitation 24 hours prior	Precipitation during application	Precipitation 24 hours following
01/30/2023	Surface (irrigation)	No precipitation	No precipitation	No precipitation

Source description	Material type	N (lbs/acre)	P (lbs/acre)	K (lbs/acre)	Salt (lbs/acre)	Amount
Lagoon Water	Process wastewater	12.50	2.11	13.42	75.81	142,408.00 *gal*
Application event totals		12.50	2.11	13.42	75.81	

Application date	Application method	Precipitation 24 hours prior	Precipitation during application	Precipitation 24 hours following
02/06/2023	Surface (irrigation)	No precipitation	No precipitation	No precipitation

Source description	Material type	N (lbs/acre)	P (lbs/acre)	K (lbs/acre)	Salt (lbs/acre)	Amount
Lagoon Water	Process wastewater	3.72	0.63	4.00	22.58	42,408.00 *gal*
Application event totals		3.72	0.63	4.00	22.58	

6A - 05/15/2023: Corn, silage

Field name: 6A

Crop: Corn, silage

Plant date: 05/15/2023

Application date	Application method	Precipitation 24 hours prior	Precipitation during application	Precipitation 24 hours following
04/27/2023	Surface (irrigation)	No precipitation	No precipitation	No precipitation

Source description	Material type	N (lbs/acre)	P (lbs/acre)	K (lbs/acre)	Salt (lbs/acre)	Amount
Lagoon Water	Process wastewater	38.06	12.27	53.02	326.26	2,362,050.00 *gal*
Canal Water	Surface water	0.58	0.00	0.00	96.29	7,873,500.00 *gal*
Application event totals		38.64	12.27	53.02	422.55	

6A - 05/15/2023: Corn, silage

Application date	Application method	Precipitation 24 hours prior	Precipitation during application	Precipitation 24 hours following
06/15/2023	Surface (irrigation)	No precipitation	No precipitation	No precipitation

Source description	Material type	N (lbs/acre)	P (lbs/acre)	K (lbs/acre)	Salt (lbs/acre)	Amount
Lagoon Water	Process wastewater	38.06	12.27	53.02	326.26	2,362,050.00 *gal*
Canal Water	Surface water	0.58	0.00	0.00	96.29	7,873,500.00 *gal*
Application event totals		38.64	12.27	53.02	422.55	

06/28/2023	Surface (irrigation)	No precipitation	No precipitation	No precipitation

Source description	Material type	N (lbs/acre)	P (lbs/acre)	K (lbs/acre)	Salt (lbs/acre)	Amount
Lagoon Water	Process wastewater	20.30	6.54	28.28	174.00	1,259,760.00 *gal*
Canal Water	Surface water	0.40	0.00	0.00	67.40	5,511,450.00 *gal*
Application event totals		20.70	6.54	28.28	241.41	

07/10/2023	Surface (irrigation)	No precipitation	No precipitation	No precipitation

Source description	Material type	N (lbs/acre)	P (lbs/acre)	K (lbs/acre)	Salt (lbs/acre)	Amount
Lagoon Water	Process wastewater	20.30	6.54	28.28	174.00	1,259,760.00 *gal*
Canal Water	Surface water	0.40	0.00	0.00	67.40	5,511,450.00 *gal*
Application event totals		20.70	6.54	28.28	241.41	

07/22/2023	Surface (irrigation)	No precipitation	No precipitation	No precipitation

Source description	Material type	N (lbs/acre)	P (lbs/acre)	K (lbs/acre)	Salt (lbs/acre)	Amount
Lagoon Water	Process wastewater	20.30	6.54	28.28	174.00	1,259,760.00 *gal*
Canal Water	Surface water	0.40	0.00	0.00	67.40	5,511,450.00 *gal*
Application event totals		20.70	6.54	28.28	241.41	

08/03/2023	Surface (irrigation)	No precipitation	No precipitation	No precipitation

Source description	Material type	N (lbs/acre)	P (lbs/acre)	K (lbs/acre)	Salt (lbs/acre)	Amount
Lagoon Water	Process wastewater	20.30	6.54	28.28	174.00	1,259,760.00 *gal*
Canal Water	Surface water	0.40	0.00	0.00	67.40	5,511,450.00 *gal*
Application event totals		20.70	6.54	28.28	241.41	

6A - 05/15/2023: Corn, silage

Application date	Application method	Precipitation 24 hours prior	Precipitation during application	Precipitation 24 hours following
08/15/2023	Surface (irrigation)	No precipitation	No precipitation	No precipitation

Source description	Material type	N (lbs/acre)	P (lbs/acre)	K (lbs/acre)	Salt (lbs/acre)	Amount
Lagoon Water	Process wastewater	20.30	6.54	28.28	174.00	1,259,760.00 *gal*
Canal Water	Surface water	0.40	0.00	0.00	67.40	5,511,450.00 *gal*
Application event totals		20.70	6.54	28.28	241.41	

08/27/2023	Surface (irrigation)	No precipitation	No precipitation	No precipitation

Source description	Material type	N (lbs/acre)	P (lbs/acre)	K (lbs/acre)	Salt (lbs/acre)	Amount
Lagoon Water	Process wastewater	20.30	6.54	28.28	174.00	1,259,760.00 *gal*
Canal Water	Surface water	0.40	0.00	0.00	67.40	5,511,450.00 *gal*
Application event totals		20.70	6.54	28.28	241.41	

6B - 11/01/2022: Oats, silage-soft dough

Field name: 6B

Crop: Oats, silage-soft dough

Plant date: 11/01/2022

Application date	Application method	Precipitation 24 hours prior	Precipitation during application	Precipitation 24 hours following
01/16/2023	Surface (irrigation)	No precipitation	No precipitation	No precipitation

Source description	Material type	N (lbs/acre)	P (lbs/acre)	K (lbs/acre)	Salt (lbs/acre)	Amount
Lagoon Water	Process wastewater	12.79	2.16	13.74	77.60	17,592.00 *gal*
Application event totals		12.79	2.16	13.74	77.60	

01/23/2023	Surface (irrigation)	No precipitation	No precipitation	No precipitation

Source description	Material type	N (lbs/acre)	P (lbs/acre)	K (lbs/acre)	Salt (lbs/acre)	Amount
Lagoon Water	Process wastewater	12.79	2.16	13.74	77.60	17,592.00 *gal*
Application event totals		12.79	2.16	13.74	77.60	

6B - 11/01/2022: Oats, silage-soft dough

Application date	Application method	Precipitation 24 hours prior	Precipitation during application	Precipitation 24 hours following
01/30/2023	Surface (irrigation)	No precipitation	No precipitation	No precipitation

Source description	Material type	N (lbs/acre)	P (lbs/acre)	K (lbs/acre)	Salt (lbs/acre)	Amount
Lagoon Water	Process wastewater	12.07	2.04	12.96	73.19	16,592.00 *gal*
Application event totals		12.07	2.04	12.96	73.19	

02/06/2023	Surface (irrigation)	No precipitation	No precipitation	No precipitation

Source description	Material type	N (lbs/acre)	P (lbs/acre)	K (lbs/acre)	Salt (lbs/acre)	Amount
Lagoon Water	Process wastewater	5.52	0.93	5.93	33.49	7,592.00 *gal*
Application event totals		5.52	0.93	5.93	33.49	

6B - 05/15/2023: Corn, silage

Field name: 6B

Crop: Corn, silage

Plant date: 05/15/2023

Application date	Application method	Precipitation 24 hours prior	Precipitation during application	Precipitation 24 hours following
04/27/2023	Surface (irrigation)	No precipitation	No precipitation	No precipitation

Source description	Material type	N (lbs/acre)	P (lbs/acre)	K (lbs/acre)	Salt (lbs/acre)	Amount
Lagoon Water	Process wastewater	38.06	12.27	53.02	326.26	285,075.00 *gal*
Ag Well #4	Ground water	0.06	0.00	0.00	84.96	475,125.00 *gal*
Canal Water	Surface water	0.29	0.00	0.00	48.15	475,125.00 *gal*
Application event totals		38.41	12.27	53.02	459.36	

06/15/2023	Surface (irrigation)	No precipitation	No precipitation	No precipitation

Source description	Material type	N (lbs/acre)	P (lbs/acre)	K (lbs/acre)	Salt (lbs/acre)	Amount
Lagoon Water	Process wastewater	38.06	12.27	53.02	326.26	285,075.00 *gal*
Ag Well #4	Ground water	0.06	0.00	0.00	84.96	475,125.00 *gal*
Canal Water	Surface water	0.29	0.00	0.00	48.15	475,125.00 *gal*
Application event totals		38.41	12.27	53.02	459.36	

6B - 05/15/2023: Corn, silage

Application date	Application method	Precipitation 24 hours prior	Precipitation during application	Precipitation 24 hours following
06/28/2023	Surface (irrigation)	No precipitation	No precipitation	No precipitation

Source description	Material type	N (lbs/acre)	P (lbs/acre)	K (lbs/acre)	Salt (lbs/acre)	Amount
Lagoon Water	Process wastewater	20.30	6.54	28.28	174.00	152,040.00 *gal*
Ag Well #4	Ground water	0.04	0.00	0.00	59.47	332,587.00 *gal*
Canal Water	Surface water	0.20	0.00	0.00	33.70	332,587.00 *gal*
Application event totals		20.54	6.54	28.28	267.18	

07/10/2023	Surface (irrigation)	No precipitation	No precipitation	No precipitation

Source description	Material type	N (lbs/acre)	P (lbs/acre)	K (lbs/acre)	Salt (lbs/acre)	Amount
Lagoon Water	Process wastewater	20.30	6.54	28.28	174.00	152,040.00 *gal*
Ag Well #4	Ground water	0.04	0.00	0.00	59.47	332,587.00 *gal*
Canal Water	Surface water	0.20	0.00	0.00	33.70	332,587.00 *gal*
Application event totals		20.54	6.54	28.28	267.18	

07/22/2023	Surface (irrigation)	No precipitation	No precipitation	No precipitation

Source description	Material type	N (lbs/acre)	P (lbs/acre)	K (lbs/acre)	Salt (lbs/acre)	Amount
Lagoon Water	Process wastewater	20.30	6.54	28.28	174.00	152,040.00 *gal*
Ag Well #4	Ground water	0.04	0.00	0.00	59.47	332,587.00 *gal*
Canal Water	Surface water	0.20	0.00	0.00	33.70	332,587.00 *gal*
Application event totals		20.54	6.54	28.28	267.18	

08/03/2023	Surface (irrigation)	No precipitation	No precipitation	No precipitation

Source description	Material type	N (lbs/acre)	P (lbs/acre)	K (lbs/acre)	Salt (lbs/acre)	Amount
Lagoon Water	Process wastewater	20.30	6.54	28.28	174.00	152,040.00 *gal*
Ag Well #4	Ground water	0.04	0.00	0.00	59.47	332,587.00 *gal*
Canal Water	Surface water	0.20	0.00	0.00	33.70	332,587.00 *gal*
Application event totals		20.54	6.54	28.28	267.18	

6B - 05/15/2023: Corn, silage

Application date	Application method	Precipitation 24 hours prior	Precipitation during application	Precipitation 24 hours following
08/15/2023	Surface (irrigation)	No precipitation	No precipitation	No precipitation

Source description	Material type	N (lbs/acre)	P (lbs/acre)	K (lbs/acre)	Salt (lbs/acre)	Amount
Lagoon Water	Process wastewater	20.30	6.54	28.28	174.00	152,040.00 *gal*
Ag Well #4	Ground water	0.04	0.00	0.00	59.47	332,587.00 *gal*
Canal Water	Surface water	0.20	0.00	0.00	33.70	332,587.00 *gal*
Application event totals		20.54	6.54	28.28	267.18	

08/28/2023	Surface (irrigation)	No precipitation	No precipitation	No precipitation

Source description	Material type	N (lbs/acre)	P (lbs/acre)	K (lbs/acre)	Salt (lbs/acre)	Amount
Lagoon Water	Process wastewater	20.30	6.54	28.28	174.00	152,040.00 *gal*
Ag Well #4	Ground water	0.04	0.00	0.00	59.47	332,587.00 *gal*
Canal Water	Surface water	0.20	0.00	0.00	33.70	332,587.00 *gal*
Application event totals		20.54	6.54	28.28	267.18	

7 - 11/01/2022: Oats, silage-soft dough

Field name: 7

Crop: Oats, silage-soft dough

Plant date: 11/01/2022

Application date	Application method	Precipitation 24 hours prior	Precipitation during application	Precipitation 24 hours following
12/26/2022	Surface (irrigation)	No precipitation	No precipitation	No precipitation

Source description	Material type	N (lbs/acre)	P (lbs/acre)	K (lbs/acre)	Salt (lbs/acre)	Amount
Lagoon Water	Process wastewater	68.97	11.65	74.06	418.33	420,000.00 *gal*
Application event totals		68.97	11.65	74.06	418.33	

01/02/2023	Surface (irrigation)	No precipitation	No precipitation	No precipitation

Source description	Material type	N (lbs/acre)	P (lbs/acre)	K (lbs/acre)	Salt (lbs/acre)	Amount
Lagoon Water	Process wastewater	85.39	14.42	91.69	517.93	520,000.00 *gal*
Application event totals		85.39	14.42	91.69	517.93	

7 - 05/15/2023: Corn, silage

7 - 05/15/2023: Corn, silage

Field name: 7

Crop: Corn, silage

Plant date: 05/15/2023

Application date	Application method	Precipitation 24 hours prior	Precipitation during application	Precipitation 24 hours following
04/29/2023	Surface (irrigation)	No precipitation	No precipitation	No precipitation

Source description	Material type	N (lbs/acre)	P (lbs/acre)	K (lbs/acre)	Salt (lbs/acre)	Amount
Lagoon Water	Process wastewater	7.91	2.55	11.02	67.83	262,475.00 *gal*
Ag Well #5	Ground water	0.06	0.00	0.00	21.52	2,104,125.00 *gal*
Canal Water	Surface water	0.29	0.00	0.00	48.15	2,104,125.00 *gal*
Application event totals		8.26	2.55	11.02	137.50	

06/17/2023	Surface (irrigation)	No precipitation	No precipitation	No precipitation

Source description	Material type	N (lbs/acre)	P (lbs/acre)	K (lbs/acre)	Salt (lbs/acre)	Amount
Lagoon Water	Process wastewater	7.91	2.55	11.02	67.83	262,475.00 *gal*
Ag Well #5	Ground water	0.06	0.00	0.00	21.52	2,104,125.00 *gal*
Canal Water	Surface water	0.29	0.00	0.00	48.15	2,104,125.00 *gal*
Application event totals		8.26	2.55	11.02	137.50	

06/29/2023	Surface (irrigation)	No precipitation	No precipitation	No precipitation

Source description	Material type	N (lbs/acre)	P (lbs/acre)	K (lbs/acre)	Salt (lbs/acre)	Amount
Lagoon Water	Process wastewater	20.30	6.54	28.28	174.00	673,320.00 *gal*
Ag Well #5	Ground water	0.04	0.00	0.00	15.07	1,472,887.50 *gal*
Canal Water	Surface water	0.20	0.00	0.00	33.70	1,472,887.50 *gal*
Application event totals		20.54	6.54	28.28	222.77	

07/11/2023	Surface (irrigation)	No precipitation	No precipitation	No precipitation

Source description	Material type	N (lbs/acre)	P (lbs/acre)	K (lbs/acre)	Salt (lbs/acre)	Amount
Lagoon Water	Process wastewater	20.30	6.54	28.28	174.00	673,320.00 *gal*
Ag Well #5	Ground water	0.04	0.00	0.00	15.07	1,472,887.50 *gal*
Canal Water	Surface water	0.20	0.00	0.00	33.70	1,472,887.50 *gal*
Application event totals		20.54	6.54	28.28	222.77	

7 - 05/15/2023: Corn, silage

Application date	Application method	Precipitation 24 hours prior	Precipitation during application	Precipitation 24 hours following
07/23/2023	Surface (irrigation)	No precipitation	No precipitation	No precipitation

Source description	Material type	N (lbs/acre)	P (lbs/acre)	K (lbs/acre)	Salt (lbs/acre)	Amount
Lagoon Water	Process wastewater	20.30	6.54	28.28	174.00	673,320.00 *gal*
Ag Well #5	Ground water	0.00	0.00	0.00	15.07	1,472,887.50 *gal*
Canal Water	Surface water	0.20	0.00	0.00	33.70	1,472,887.50 *gal*
Application event totals		20.50	6.54	28.28	222.77	

08/04/2023	Surface (irrigation)	No precipitation	No precipitation	No precipitation

Source description	Material type	N (lbs/acre)	P (lbs/acre)	K (lbs/acre)	Salt (lbs/acre)	Amount
Lagoon Water	Process wastewater	20.30	6.54	28.28	174.00	673,320.00 *gal*
Ag Well #5	Ground water	0.04	0.00	0.00	15.07	1,472,887.50 *gal*
Canal Water	Surface water	0.20	0.00	0.00	33.70	1,472,887.50 *gal*
Application event totals		20.54	6.54	28.28	222.77	

08/16/2023	Surface (irrigation)	No precipitation	No precipitation	No precipitation

Source description	Material type	N (lbs/acre)	P (lbs/acre)	K (lbs/acre)	Salt (lbs/acre)	Amount
Lagoon Water	Process wastewater	20.30	6.54	28.28	174.00	673,320.00 *gal*
Ag Well #5	Ground water	0.04	0.00	0.00	15.07	1,472,887.50 *gal*
Canal Water	Surface water	0.20	0.00	0.00	33.70	1,472,887.50 *gal*
Application event totals		20.54	6.54	28.28	222.77	

08/28/2023	Surface (irrigation)	No precipitation	No precipitation	No precipitation

Source description	Material type	N (lbs/acre)	P (lbs/acre)	K (lbs/acre)	Salt (lbs/acre)	Amount
Lagoon Water	Process wastewater	20.30	6.54	28.28	174.00	673,320.00 *gal*
Ag Well #5	Ground water	0.04	0.00	0.00	15.07	1,472,887.50 *gal*
Canal Water	Surface water	0.20	0.00	0.00	33.70	1,472,887.50 *gal*
Application event totals		20.54	6.54	28.28	222.77	

8 - 11/01/2022: Oats, silage-soft dough

Field name: 8

Crop: Oats, silage-soft dough

Plant date: 11/01/2022

8 - 11/01/2022: Oats, silage-soft dough

Application date	Application method	Precipitation 24 hours prior	Precipitation during application	Precipitation 24 hours following
12/19/2022	Surface (irrigation)	No precipitation	No precipitation	No precipitation

Source description	Material type	N (lbs/acre)	P (lbs/acre)	K (lbs/acre)	Salt (lbs/acre)	Amount
Lagoon Water	Process wastewater	72.72	12.28	78.09	441.09	200,000.00 *gal*
Application event totals		72.72	12.28	78.09	441.09	

8 - 05/15/2023: Corn, silage

Field name: 8

Crop: Corn, silage

Plant date: 05/15/2023

Application date	Application method	Precipitation 24 hours prior	Precipitation during application	Precipitation 24 hours following
04/30/2023	Surface (irrigation)	No precipitation	No precipitation	No precipitation

Source description	Material type	N (lbs/acre)	P (lbs/acre)	K (lbs/acre)	Salt (lbs/acre)	Amount
Lagoon Water	Process wastewater	38.06	12.27	53.02	326.26	570,150.00 *gal*
Ag Well #5	Ground water	0.06	0.00	0.00	21.52	950,250.00 *gal*
Canal Water	Surface water	0.29	0.00	0.00	48.15	950,250.00 *gal*
Application event totals		38.41	12.27	53.02	395.93	

06/18/2023	Surface (irrigation)	No precipitation	No precipitation	No precipitation

Source description	Material type	N (lbs/acre)	P (lbs/acre)	K (lbs/acre)	Salt (lbs/acre)	Amount
Lagoon Water	Process wastewater	38.06	12.27	53.02	326.26	570,150.00 *gal*
Ag Well #5	Ground water	0.06	0.00	0.00	21.52	950,250.00 *gal*
Canal Water	Surface water	0.29	0.00	0.00	48.15	950,250.00 *gal*
Application event totals		38.41	12.27	53.02	395.93	

06/30/2023	Surface (irrigation)	No precipitation	No precipitation	No precipitation

Source description	Material type	N (lbs/acre)	P (lbs/acre)	K (lbs/acre)	Salt (lbs/acre)	Amount
Lagoon Water	Process wastewater	20.30	6.54	28.28	174.00	304,080.00 *gal*
Ag Well #5	Ground water	0.04	0.00	0.00	15.07	665,175.00 *gal*
Canal Water	Surface water	0.20	0.00	0.00	33.70	665,175.00 *gal*
Application event totals		20.54	6.54	28.28	222.77	

8 - 05/15/2023: Corn, silage

Application date	Application method	Precipitation 24 hours prior	Precipitation during application	Precipitation 24 hours following
07/12/2023	Surface (irrigation)	No precipitation	No precipitation	No precipitation

Source description	Material type	N (lbs/acre)	P (lbs/acre)	K (lbs/acre)	Salt (lbs/acre)	Amount
Lagoon Water	Process wastewater	20.30	6.54	28.28	174.00	304,080.00 *gal*
Ag Well #5	Ground water	0.04	0.00	0.00	15.07	665,175.00 *gal*
Canal Water	Surface water	0.20	0.00	0.00	33.70	665,175.00 *gal*
Application event totals		20.54	6.54	28.28	222.77	

07/24/2023	Surface (irrigation)	No precipitation	No precipitation	No precipitation

Source description	Material type	N (lbs/acre)	P (lbs/acre)	K (lbs/acre)	Salt (lbs/acre)	Amount
Lagoon Water	Process wastewater	20.30	6.54	28.28	174.00	304,080.00 *gal*
Ag Well #5	Ground water	0.04	0.00	0.00	15.07	665,175.00 *gal*
Canal Water	Surface water	0.20	0.00	0.00	33.70	665,175.00 *gal*
Application event totals		20.54	6.54	28.28	222.77	

08/05/2023	Surface (irrigation)	No precipitation	No precipitation	No precipitation

Source description	Material type	N (lbs/acre)	P (lbs/acre)	K (lbs/acre)	Salt (lbs/acre)	Amount
Lagoon Water	Process wastewater	20.30	6.54	28.28	174.00	304,080.00 *gal*
Ag Well #5	Ground water	0.04	0.00	0.00	15.07	665,175.00 *gal*
Canal Water	Surface water	0.20	0.00	0.00	33.70	665,175.00 *gal*
Application event totals		20.54	6.54	28.28	222.77	

08/17/2023	Surface (irrigation)	No precipitation	No precipitation	No precipitation

Source description	Material type	N (lbs/acre)	P (lbs/acre)	K (lbs/acre)	Salt (lbs/acre)	Amount
Lagoon Water	Process wastewater	20.30	6.54	28.28	174.00	304,080.00 *gal*
Ag Well #5	Ground water	0.00	0.00	0.00	15.07	665,175.00 *gal*
Canal Water	Surface water	0.20	0.00	0.00	33.70	665,175.00 *gal*
Application event totals		20.50	6.54	28.28	222.77	

8 - 05/15/2023: Corn, silage

Application date	Application method	Precipitation 24 hours prior	Precipitation during application	Precipitation 24 hours following
08/29/2023	Surface (irrigation)	No precipitation	No precipitation	No precipitation

Source description	Material type	N (lbs/acre)	P (lbs/acre)	K (lbs/acre)	Salt (lbs/acre)	Amount
Lagoon Water	Process wastewater	20.30	6.54	28.28	174.00	304,080.00 *gal*
Ag Well #5	Ground water	0.00	0.00	0.00	15.07	665,175.00 *gal*
Canal Water	Surface water	0.20	0.00	0.00	33.70	665,175.00 *gal*
Application event totals		20.50	6.54	28.28	222.77	

9 - 11/01/2022: Oats, silage-soft dough

Field name: 9

Crop: Oats, silage-soft dough Plant date: 11/01/2022

Application date	Application method	Precipitation 24 hours prior	Precipitation during application	Precipitation 24 hours following
01/09/2023	Surface (irrigation)	No precipitation	No precipitation	No precipitation

Source description	Material type	N (lbs/acre)	P (lbs/acre)	K (lbs/acre)	Salt (lbs/acre)	Amount
Lagoon Water	Process wastewater	33.94	5.73	36.44	205.84	120,000.00 *gal*
Application event totals		33.94	5.73	36.44	205.84	

9 - 05/15/2023: Corn, silage

Field name: 9

Crop: Corn, silage Plant date: 05/15/2023

Application date	Application method	Precipitation 24 hours prior	Precipitation during application	Precipitation 24 hours following
05/01/2023	Surface (irrigation)	No precipitation	No precipitation	No precipitation

Source description	Material type	N (lbs/acre)	P (lbs/acre)	K (lbs/acre)	Salt (lbs/acre)	Amount
Lagoon Water	Process wastewater	38.06	12.27	53.02	326.26	733,050.00 *gal*
Ag Well #4	Ground water	0.00	0.00	0.00	84.96	1,221,750.00 *gal*
Canal Water	Surface water	0.29	0.00	0.00	48.15	1,221,750.00 *gal*
Application event totals		38.35	12.27	53.02	459.36	

9 - 05/15/2023: Corn, silage

Application date	Application method	Precipitation 24 hours prior	Precipitation during application	Precipitation 24 hours following
06/19/2023	Surface (irrigation)	No precipitation	No precipitation	No precipitation

Source description	Material type	N (lbs/acre)	P (lbs/acre)	K (lbs/acre)	Salt (lbs/acre)	Amount
Lagoon Water	Process wastewater	38.06	12.27	53.02	326.26	733,050.00 *gal*
Ag Well #4	Ground water	0.00	0.00	0.00	84.96	1,221,750.00 *gal*
Canal Water	Surface water	0.29	0.00	0.00	48.15	1,221,750.00 *gal*
Application event totals		38.35	12.27	53.02	459.36	

06/30/2023	Surface (irrigation)	No precipitation	No precipitation	No precipitation

Source description	Material type	N (lbs/acre)	P (lbs/acre)	K (lbs/acre)	Salt (lbs/acre)	Amount
Lagoon Water	Process wastewater	20.25	6.53	28.21	173.58	390,000.00 *gal*
Ag Well #4	Ground water	0.00	0.00	0.00	59.48	855,255.00 *gal*
Canal Water	Surface water	0.20	0.00	0.00	33.70	855,255.00 *gal*
Application event totals		20.45	6.53	28.21	266.75	

07/13/2023	Surface (irrigation)	No precipitation	No precipitation	No precipitation

Source description	Material type	N (lbs/acre)	P (lbs/acre)	K (lbs/acre)	Salt (lbs/acre)	Amount
Lagoon Water	Process wastewater	20.25	6.53	28.21	173.58	390,000.00 *gal*
Ag Well #4	Ground water	0.00	0.00	0.00	59.48	855,255.00 *gal*
Canal Water	Surface water	0.20	0.00	0.00	33.70	855,255.00 *gal*
Application event totals		20.45	6.53	28.21	266.75	

07/24/2023	Surface (irrigation)	No precipitation	No precipitation	No precipitation

Source description	Material type	N (lbs/acre)	P (lbs/acre)	K (lbs/acre)	Salt (lbs/acre)	Amount
Lagoon Water	Process wastewater	20.25	6.53	28.21	173.58	390,000.00 *gal*
Ag Well #4	Ground water	0.00	0.00	0.00	59.48	855,255.00 *gal*
Canal Water	Surface water	0.20	0.00	0.00	33.70	855,255.00 *gal*
Application event totals		20.45	6.53	28.21	266.75	

9 - 05/15/2023: Corn, silage

Application date	Application method	Precipitation 24 hours prior	Precipitation during application	Precipitation 24 hours following
08/05/2023	Surface (irrigation)	No precipitation	No precipitation	No precipitation

Source description	Material type	N (lbs/acre)	P (lbs/acre)	K (lbs/acre)	Salt (lbs/acre)	Amount
Lagoon Water	Process wastewater	20.25	6.53	28.21	173.58	390,000.00 *gal*
Ag Well #4	Ground water	0.00	0.00	0.00	59.48	855,255.00 *gal*
Canal Water	Surface water	0.20	0.00	0.00	33.70	855,255.00 *gal*
Application event totals		20.45	6.53	28.21	266.75	

08/17/2023	Surface (irrigation)	No precipitation	No precipitation	No precipitation

Source description	Material type	N (lbs/acre)	P (lbs/acre)	K (lbs/acre)	Salt (lbs/acre)	Amount
Lagoon Water	Process wastewater	20.25	6.53	28.21	173.58	390,000.00 *gal*
Ag Well #4	Ground water	0.00	0.00	0.00	59.48	855,255.00 *gal*
Canal Water	Surface water	0.20	0.00	0.00	33.70	855,255.00 *gal*
Application event totals		20.45	6.53	28.21	266.75	

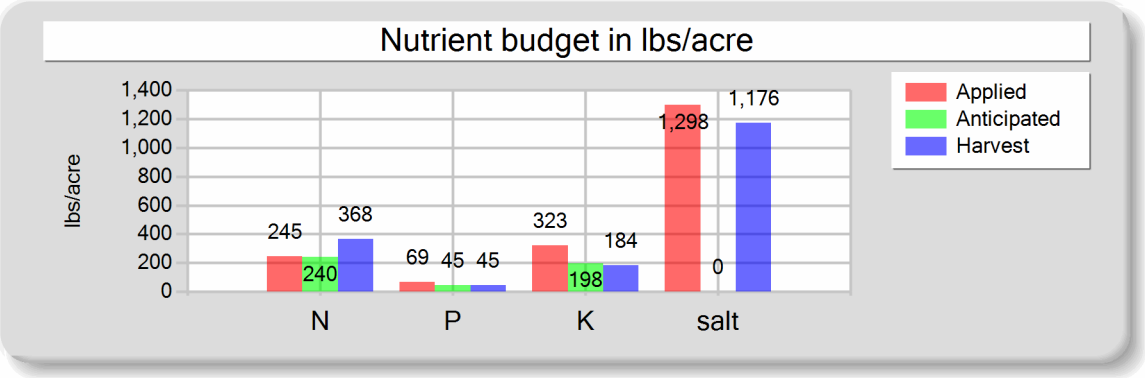
08/29/2023	Surface (irrigation)	No precipitation	No precipitation	No precipitation

Source description	Material type	N (lbs/acre)	P (lbs/acre)	K (lbs/acre)	Salt (lbs/acre)	Amount
Lagoon Water	Process wastewater	20.25	6.53	28.21	173.58	390,000.00 *gal*
Ag Well #4	Ground water	0.00	0.00	0.00	59.48	855,255.00 *gal*
Canal Water	Surface water	0.20	0.00	0.00	33.70	855,255.00 *gal*
Application event totals		20.45	6.53	28.21	266.75	

B. NUTRIENT BUDGET

1 - 05/15/2023: Corn, silage

Field name: 1 Crop: Corn, silage Plant date: 05/15/2023



	Total N (lbs/acre)	Total P (lbs/acre)	Total K (lbs/acre)	Total salt (lbs/acre)
Existing soil nutrient content	0.00	0.00	0.00	0.00
Plowdown credit	0.00	0.00	0.00	0.00
Commercial fertilizer / Other	0.00	0.00	0.00	0.00
Dry manure	0.00	0.00	0.00	0.00
Process wastewater	227.65	69.07	322.81	700.54
Fresh water	3.58	0.00	0.00	597.00
Atmospheric deposition	14.00	0.00	0.00	0.00
Total nutrients applied	245.24	69.07	322.81	1,297.55
Anticipated crop nutrient removal	240.00	45.00	198.00	0.00
Actual crop nutrient removal	368.23	44.77	183.90	1,175.82
Nutrient balance	-122.99	24.30	138.91	121.73
Applied to removed ratio	0.67	1.54	1.76	1.10

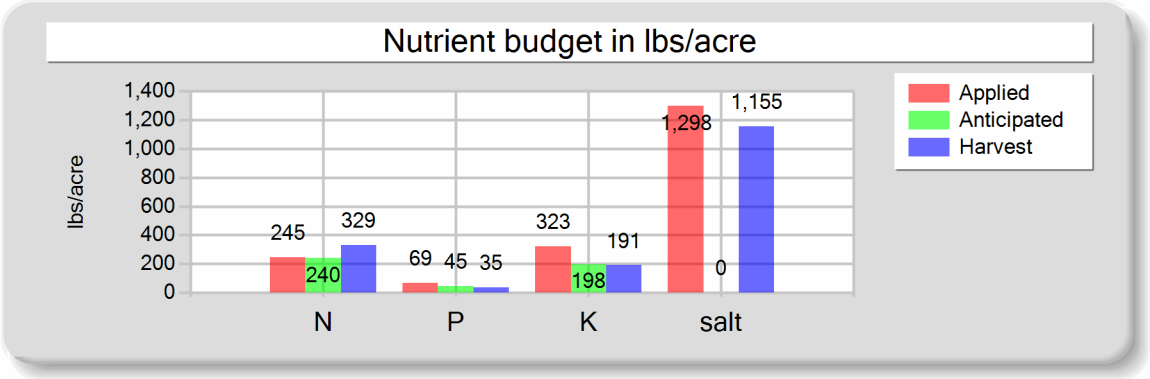
Fresh water applied	
27,774,450.00	*gallons*
1,022.84	*acre-inches*
31.00	*inches/acre*

Process wastewater applied	
6,988,410.00	*gallons*
257.36	*acre-inches*
7.80	*inches/acre*

Total harvests for the crop	
1	*harvests*

2A - 05/15/2023: Corn, silage

Field name: 2A Crop: Corn, silage Plant date: 05/15/2023



	Total N (lbs/acre)	Total P (lbs/acre)	Total K (lbs/acre)	Total salt (lbs/acre)
Existing soil nutrient content	0.00	0.00	0.00	0.00
Plowdown credit	0.00	0.00	0.00	0.00
Commercial fertilizer / Other	0.00	0.00	0.00	0.00
Dry manure	0.00	0.00	0.00	0.00
Process wastewater	227.65	69.07	322.81	700.54
Fresh water	3.58	0.00	0.00	597.00
Atmospheric deposition	14.00	0.00	0.00	0.00
Total nutrients applied	245.24	69.07	322.81	1,297.55
Anticipated crop nutrient removal	240.00	45.00	198.00	0.00
Actual crop nutrient removal	329.38	35.38	190.71	1,154.86
Nutrient balance	-84.14	33.69	132.10	142.69
Applied to removed ratio	0.74	1.95	1.69	1.12

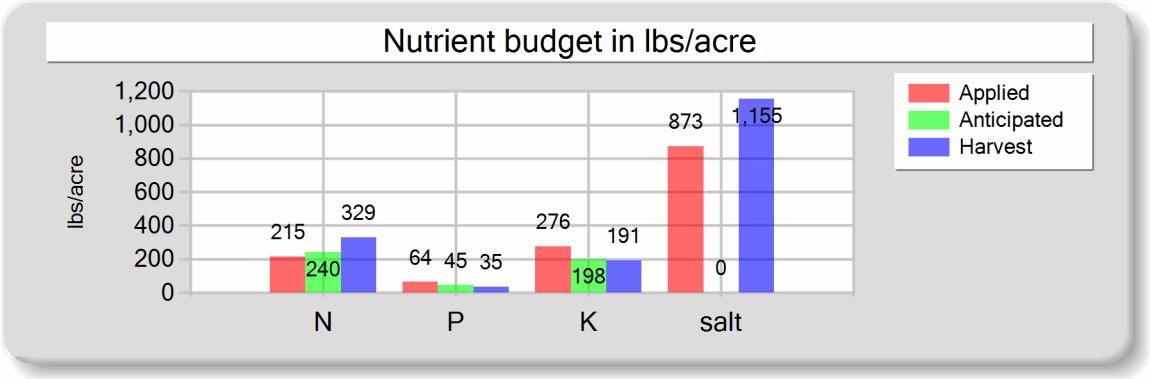
Fresh water applied	
32,824,350.00	*gallons*
1,208.81	*acre-inches*
31.00	*inches/acre*

Process wastewater applied	
8,259,030.00	*gallons*
304.15	*acre-inches*
7.80	*inches/acre*

Total harvests for the crop	
1	*harvests*

2B - 05/15/2023: Corn, silage

Field name: 2B | Crop: Corn, silage | Plant date: 05/15/2023



	Total N (lbs/acre)	Total P (lbs/acre)	Total K (lbs/acre)	Total salt (lbs/acre)
Existing soil nutrient content	0.00	0.00	0.00	0.00
Plowdown credit	0.00	0.00	0.00	0.00
Commercial fertilizer / Other	0.00	0.00	0.00	0.00
Dry manure	0.00	0.00	0.00	0.00
Process wastewater	197.82	63.76	275.53	275.53
Fresh water	3.58	0.00	0.00	597.00
Atmospheric deposition	14.00	0.00	0.00	0.00
Total nutrients applied	215.40	63.76	275.53	872.53
Anticipated crop nutrient removal	240.00	45.00	198.00	0.00
Actual crop nutrient removal	329.38	35.38	190.71	1,154.86
Nutrient balance	-113.98	28.38	84.82	-282.32
Applied to removed ratio	0.65	1.80	1.44	0.76

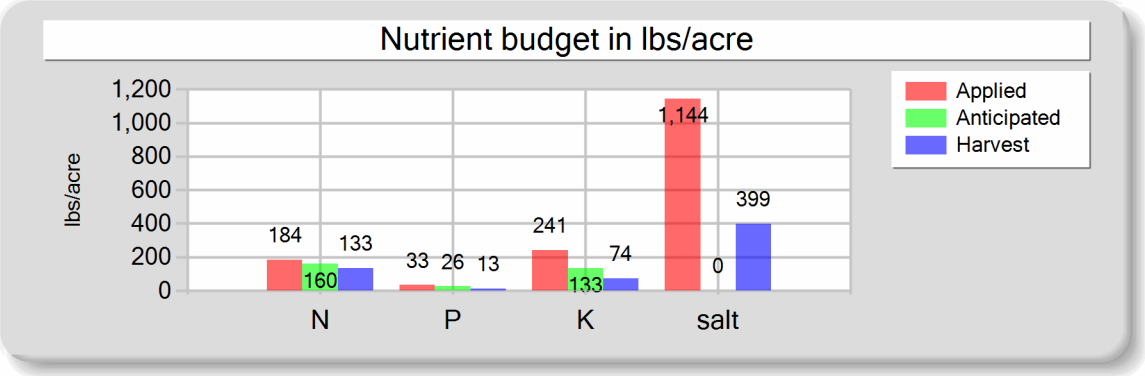
Fresh water applied	
4,208,250.00	*gallons*
154.98	*acre-inches*
31.00	*inches/acre*

Process wastewater applied	
1,058,250.00	*gallons*
38.97	*acre-inches*
7.79	*inches/acre*

Total harvests for the crop	
1	*harvests*

3 - 11/01/2022: Oats, silage-soft dough

Field name: 3 Crop: Oats, silage-soft dough Plant date: 11/01/2022



	Total N (lbs/acre)	Total P (lbs/acre)	Total K (lbs/acre)	Total salt (lbs/acre)
Existing soil nutrient content	0.00	0.00	0.00	0.00
Plowdown credit	0.00	0.00	0.00	0.00
Commercial fertilizer / Other	0.00	0.00	0.00	0.00
Dry manure	0.00	0.00	0.00	0.00
Process wastewater	177.07	32.90	240.77	1,144.02
Fresh water	0.00	0.00	0.00	0.00
Atmospheric deposition	7.00	0.00	0.00	0.00
Total nutrients applied	184.07	32.90	240.77	1,144.02
Anticipated crop nutrient removal	160.00	25.60	132.80	0.00
Actual crop nutrient removal	133.11	13.31	73.92	399.34
Nutrient balance	50.96	19.58	166.85	744.68
Applied to removed ratio	1.38	2.47	3.26	2.86

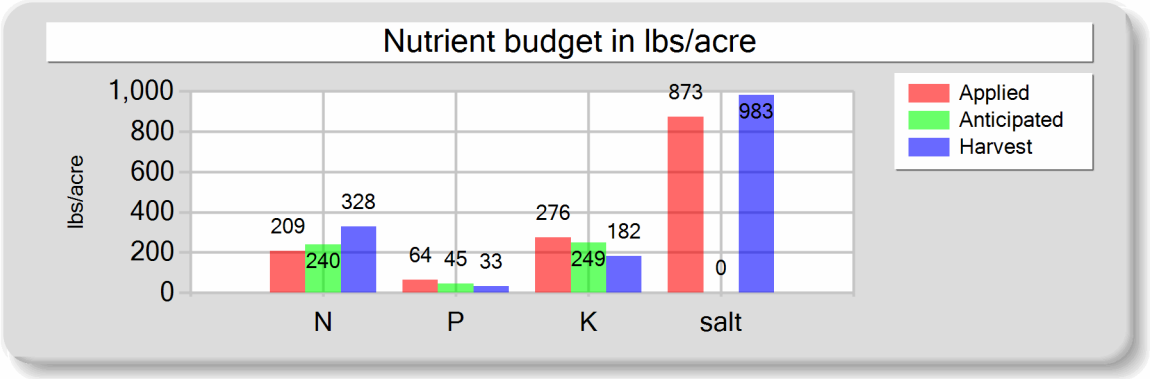
Fresh water applied	
0.00	*gallons*
0.00	*acre-inches*
0.00	*inches/acre*

Process wastewater applied	
1,340,000.00	*gallons*
49.35	*acre-inches*
1.50	*inches/acre*

Total harvests for the crop	
1	*harvests*

3 - 05/15/2023: Corn, silage

Field name: 3 Crop: Corn, silage Plant date: 05/15/2023



	Total N (lbs/acre)	Total P (lbs/acre)	Total K (lbs/acre)	Total salt (lbs/acre)
Existing soil nutrient content	0.00	0.00	0.00	0.00
Plowdown credit	0.00	0.00	0.00	0.00
Commercial fertilizer / Other	0.00	0.00	0.00	0.00
Dry manure	0.00	0.00	0.00	0.00
Process wastewater	197.93	63.80	275.69	275.69
Fresh water	3.58	0.00	0.00	597.00
Atmospheric deposition	7.00	0.00	0.00	0.00
Total nutrients applied	208.51	63.80	275.69	872.69
Anticipated crop nutrient removal	240.00	45.00	249.00	0.00
Actual crop nutrient removal	327.67	32.77	181.95	983.00
Nutrient balance	-119.16	31.03	93.74	-110.31
Applied to removed ratio	0.64	1.95	1.52	0.89

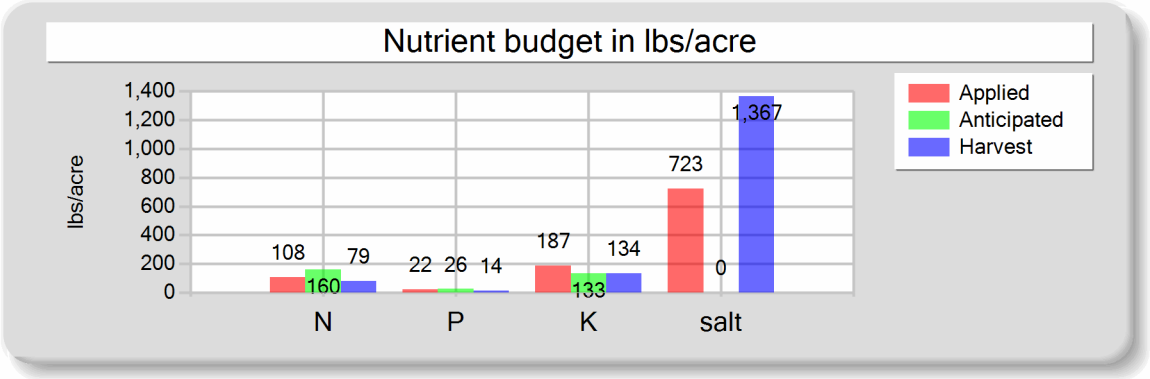
Fresh water applied	
27,774,450.00	*gallons*
1,022.84	*acre-inches*
31.00	*inches/acre*

Process wastewater applied	
6,988,410.00	*gallons*
257.36	*acre-inches*
7.80	*inches/acre*

Total harvests for the crop	
1	*harvests*

4 - 11/01/2022: Oats, silage-soft dough

Field name: 4 Crop: Oats, silage-soft dough Plant date: 11/01/2022



	Total N (lbs/acre)	Total P (lbs/acre)	Total K (lbs/acre)	Total salt (lbs/acre)
Existing soil nutrient content	0.00	0.00	0.00	0.00
Plowdown credit	0.00	0.00	0.00	0.00
Commercial fertilizer / Other	0.00	0.00	0.00	0.00
Dry manure	0.00	0.00	0.00	0.00
Process wastewater	101.49	21.75	186.84	723.23
Fresh water	0.00	0.00	0.00	0.00
Atmospheric deposition	7.00	0.00	0.00	0.00
Total nutrients applied	108.49	21.75	186.84	723.23
Anticipated crop nutrient removal	160.00	25.60	132.80	0.00
Actual crop nutrient removal	79.48	13.56	134.49	1,366.98
Nutrient balance	29.01	8.19	52.34	-643.74
Applied to removed ratio	1.36	1.60	1.39	0.53

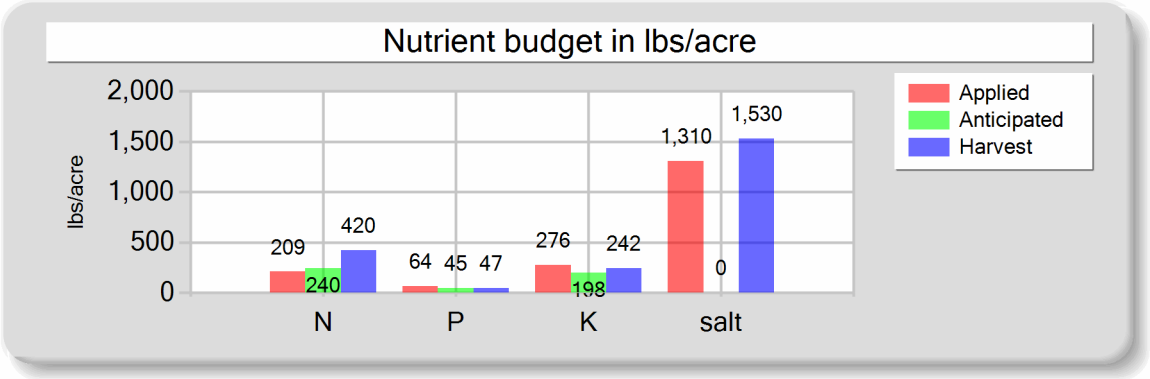
Fresh water applied
0.00 *gallons*
0.00 *acre-inches*
0.00 *inches/acre*

Process wastewater applied
1,040,000.00 *gallons*
38.30 *acre-inches*
1.06 *inches/acre*

Total harvests for the crop
1 *harvests*

4 - 05/15/2023: Corn, silage

Field name: 4 Crop: Corn, silage Plant date: 05/15/2023



	Total N (lbs/acre)	Total P (lbs/acre)	Total K (lbs/acre)	Total salt (lbs/acre)
Existing soil nutrient content	0.00	0.00	0.00	0.00
Plowdown credit	0.00	0.00	0.00	0.00
Commercial fertilizer / Other	0.00	0.00	0.00	0.00
Dry manure	0.00	0.00	0.00	0.00
Process wastewater	197.93	63.80	275.69	712.87
Fresh water	3.58	0.00	0.00	597.00
Atmospheric deposition	7.00	0.00	0.00	0.00
Total nutrients applied	208.51	63.80	275.69	1,309.87
Anticipated crop nutrient removal	240.00	45.00	198.00	0.00
Actual crop nutrient removal	419.68	47.04	242.05	1,530.37
Nutrient balance	-211.17	16.75	33.64	-220.49
Applied to removed ratio	0.50	1.36	1.14	0.86

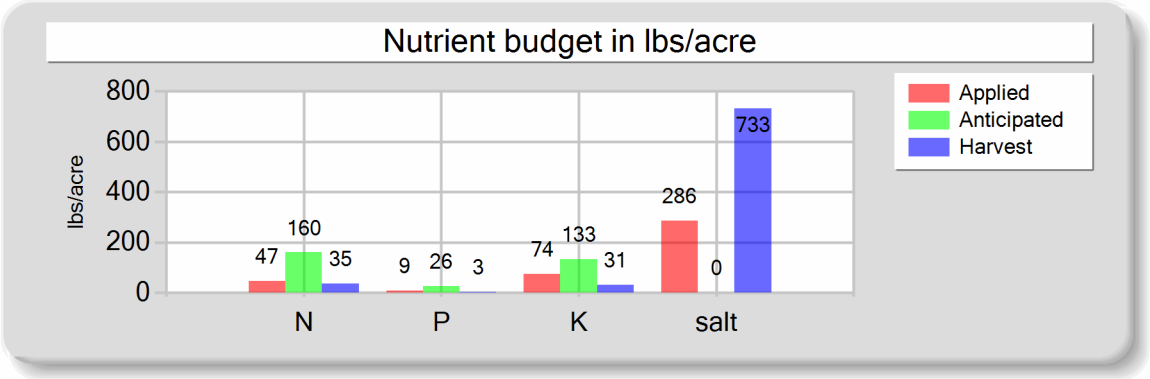
Fresh water applied	
30,299,400.00	*gallons*
1,115.82	*acre-inches*
31.00	*inches/acre*

Process wastewater applied	
7,623,720.00	*gallons*
280.76	*acre-inches*
7.80	*inches/acre*

Total harvests for the crop	
1	*harvests*

5A - 11/01/2022: Oats, silage-soft dough

Field name: 5A | Crop: Oats, silage-soft dough | Plant date: 11/01/2022



	Total N (lbs/acre)	Total P (lbs/acre)	Total K (lbs/acre)	Total salt (lbs/acre)
Existing soil nutrient content	0.00	0.00	0.00	0.00
Plowdown credit	0.00	0.00	0.00	0.00
Commercial fertilizer / Other	0.00	0.00	0.00	0.00
Dry manure	0.00	0.00	0.00	0.00
Process wastewater	40.15	8.60	73.91	286.11
Fresh water	0.00	0.00	0.00	0.00
Atmospheric deposition	7.00	0.00	0.00	0.00
Total nutrients applied	47.15	8.60	73.91	286.11
Anticipated crop nutrient removal	160.00	25.60	132.80	0.00
Actual crop nutrient removal	35.30	3.20	30.69	732.73
Nutrient balance	11.85	5.40	43.22	-446.62
Applied to removed ratio	1.34	2.69	2.41	0.39

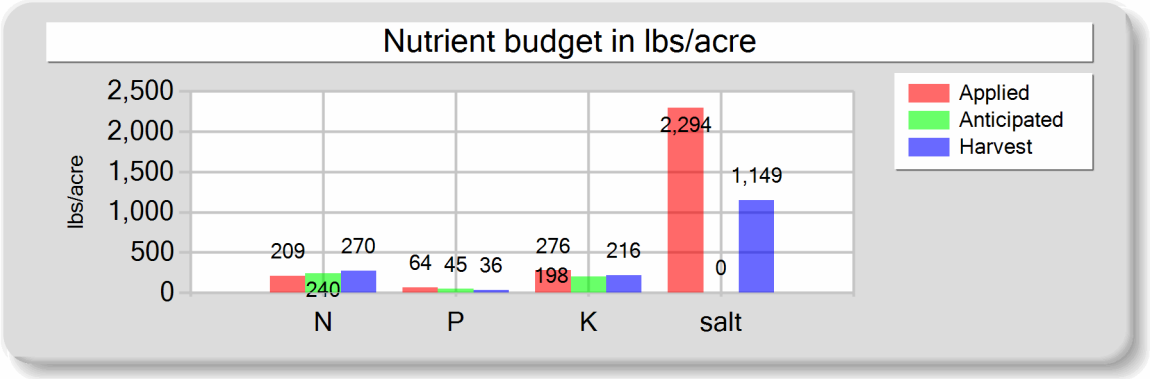
Fresh water applied
0.00 *gallons*
0.00 *acre-inches*
0.00 *inches/acre*

Process wastewater applied
160,000.00 *gallons*
5.89 *acre-inches*
0.42 *inches/acre*

Total harvests for the crop
1 *harvests*

5A - 05/15/2023: Corn, silage

Field name: 5A | Crop: Corn, silage | Plant date: 05/15/2023



	Total N (lbs/acre)	Total P (lbs/acre)	Total K (lbs/acre)	Total salt (lbs/acre)
Existing soil nutrient content	0.00	0.00	0.00	0.00
Plowdown credit	0.00	0.00	0.00	0.00
Commercial fertilizer / Other	0.00	0.00	0.00	0.00
Dry manure	0.00	0.00	0.00	0.00
Process wastewater	197.93	63.80	275.69	1,696.53
Fresh water	3.58	0.00	0.00	597.00
Atmospheric deposition	7.00	0.00	0.00	0.00
Total nutrients applied	208.51	63.80	275.69	2,293.54
Anticipated crop nutrient removal	240.00	45.00	198.00	0.00
Actual crop nutrient removal	269.95	35.70	215.96	1,149.46
Nutrient balance	-61.44	28.09	59.73	1,144.08
Applied to removed ratio	0.77	1.79	1.28	2.00

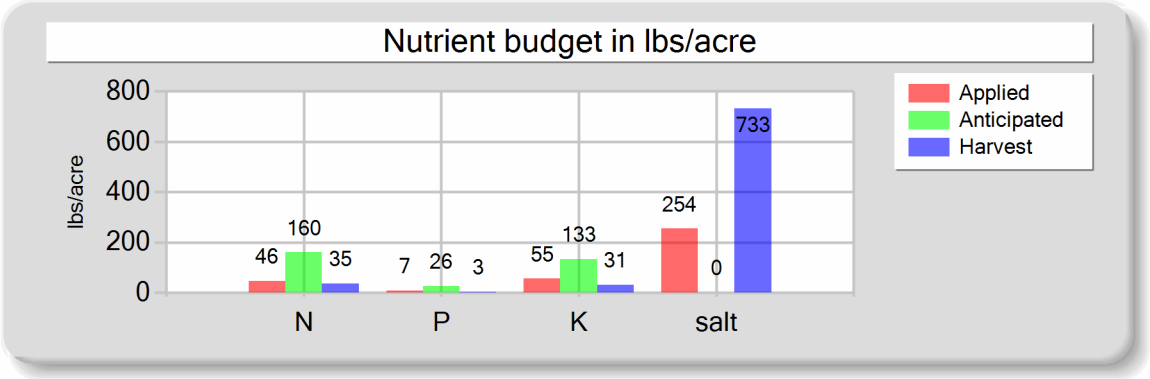
Fresh water applied	
11,783,100.00	*gallons*
433.93	*acre-inches*
31.00	*inches/acre*

Process wastewater applied	
2,964,780.00	*gallons*
109.18	*acre-inches*
7.80	*inches/acre*

Total harvests for the crop	
1	*harvests*

5B - 11/01/2022: Oats, silage-soft dough

Field name: 5B | Crop: Oats, silage-soft dough | Plant date: 11/01/2022



	Total N (lbs/acre)	Total P (lbs/acre)	Total K (lbs/acre)	Total salt (lbs/acre)
Existing soil nutrient content	0.00	0.00	0.00	0.00
Plowdown credit	0.00	0.00	0.00	0.00
Commercial fertilizer / Other	0.00	0.00	0.00	0.00
Dry manure	0.00	0.00	0.00	0.00
Process wastewater	38.78	7.34	55.11	253.83
Fresh water	0.00	0.00	0.00	0.00
Atmospheric deposition	7.00	0.00	0.00	0.00
Total nutrients applied	45.78	7.34	55.11	253.83
Anticipated crop nutrient removal	160.00	25.60	132.80	0.00
Actual crop nutrient removal	35.30	3.20	30.69	732.73
Nutrient balance	10.47	4.14	24.42	-478.90
Applied to removed ratio	1.30	2.29	1.80	0.35

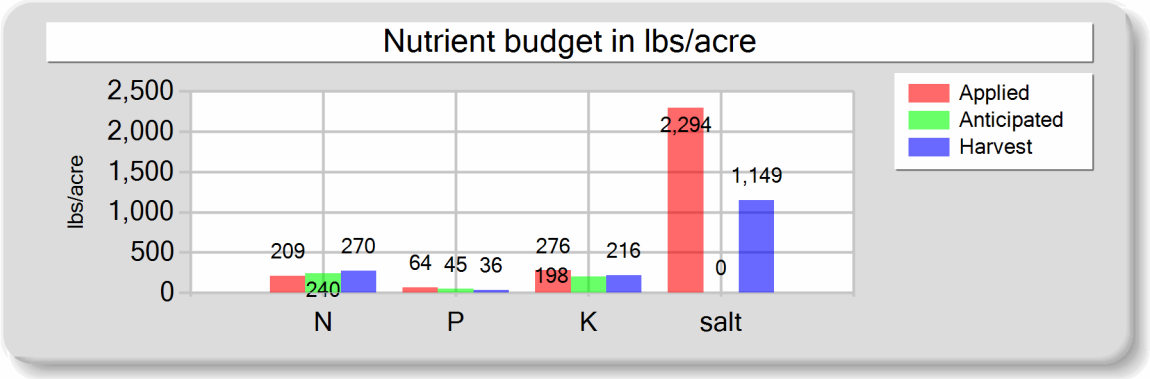
Fresh water applied
0.00 *gallons*
0.00 *acre-inches*
0.00 *inches/acre*

Process wastewater applied
110,000.00 *gallons*
4.05 *acre-inches*
0.34 *inches/acre*

Total harvests for the crop
1 *harvests*

5B - 05/15/2023: Corn, silage

Field name: 5B | Crop: Corn, silage | Plant date: 05/15/2023



	Total N (lbs/acre)	Total P (lbs/acre)	Total K (lbs/acre)	Total salt (lbs/acre)
Existing soil nutrient content	0.00	0.00	0.00	0.00
Plowdown credit	0.00	0.00	0.00	0.00
Commercial fertilizer / Other	0.00	0.00	0.00	0.00
Dry manure	0.00	0.00	0.00	0.00
Process wastewater	197.93	63.80	275.69	1,696.53
Fresh water	3.58	0.00	0.00	597.00
Atmospheric deposition	7.00	0.00	0.00	0.00
Total nutrients applied	208.51	63.80	275.69	2,293.54
Anticipated crop nutrient removal	240.00	45.00	198.00	0.00
Actual crop nutrient removal	269.95	35.70	215.96	1,149.46
Nutrient balance	-61.44	28.09	59.73	1,144.08
Applied to removed ratio	0.77	1.79	1.28	2.00

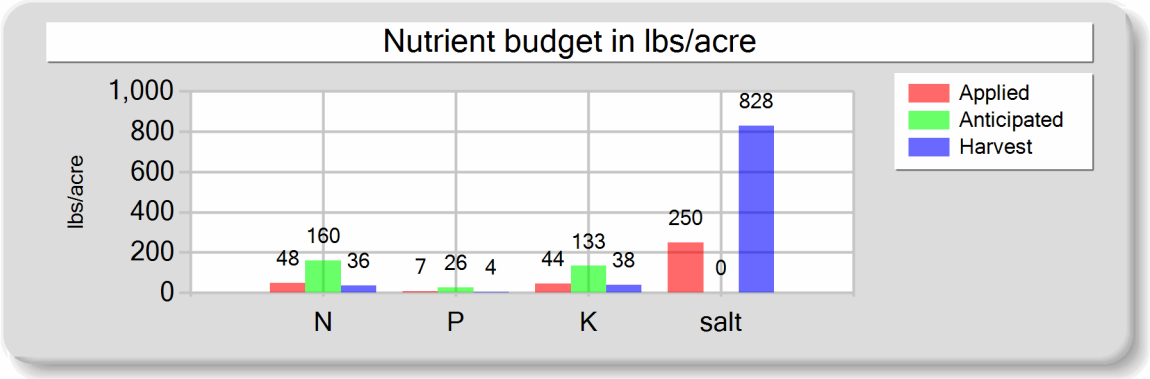
Fresh water applied	
10,099,800.00	*gallons*
371.94	*acre-inches*
31.00	*inches/acre*

Process wastewater applied	
2,541,240.00	*gallons*
93.59	*acre-inches*
7.80	*inches/acre*

Total harvests for the crop	
1	*harvests*

6A - 11/01/2022: Oats, silage-soft dough

Field name: 6A | Crop: Oats, silage-soft dough | Plant date: 11/01/2022



	Total N (lbs/acre)	Total P (lbs/acre)	Total K (lbs/acre)	Total salt (lbs/acre)
Existing soil nutrient content	0.00	0.00	0.00	0.00
Plowdown credit	0.00	0.00	0.00	0.00
Commercial fertilizer / Other	0.00	0.00	0.00	0.00
Dry manure	0.00	0.00	0.00	0.00
Process wastewater	41.22	6.96	44.26	250.01
Fresh water	0.00	0.00	0.00	0.00
Atmospheric deposition	7.00	0.00	0.00	0.00
Total nutrients applied	48.22	6.96	44.26	250.01
Anticipated crop nutrient removal	160.00	25.60	132.80	0.00
Actual crop nutrient removal	36.21	4.08	37.76	828.12
Nutrient balance	12.01	2.88	6.50	-578.11
Applied to removed ratio	1.33	1.70	1.17	0.30

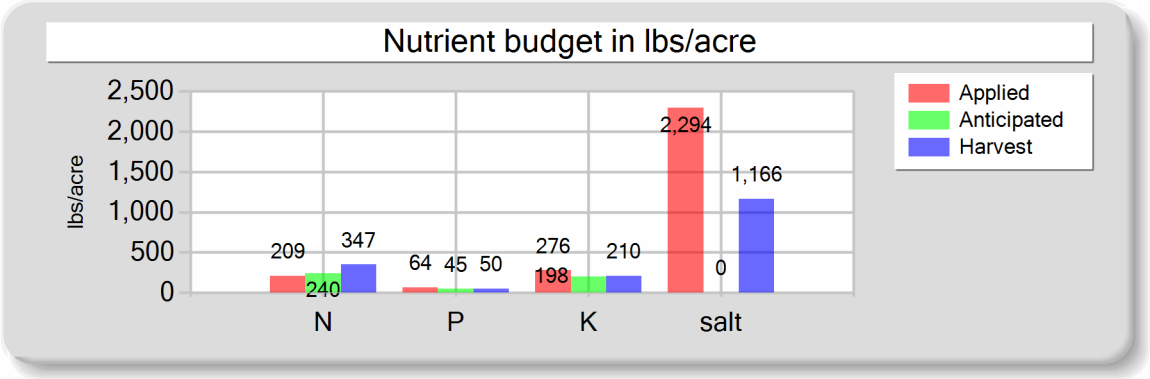
Fresh water applied	
0.00	*gallons*
0.00	*acre-inches*
0.00	*inches/acre*

Process wastewater applied	
469,632.00	*gallons*
17.29	*acre-inches*
0.30	*inches/acre*

Total harvests for the crop	
1	*harvests*

6A - 05/15/2023: Corn, silage

Field name: 6A Crop: Corn, silage Plant date: 05/15/2023



	Total N (lbs/acre)	Total P (lbs/acre)	Total K (lbs/acre)	Total salt (lbs/acre)
Existing soil nutrient content	0.00	0.00	0.00	0.00
Plowdown credit	0.00	0.00	0.00	0.00
Commercial fertilizer / Other	0.00	0.00	0.00	0.00
Dry manure	0.00	0.00	0.00	0.00
Process wastewater	197.93	63.80	275.69	1,696.53
Fresh water	3.58	0.00	0.00	597.00
Atmospheric deposition	7.00	0.00	0.00	0.00
Total nutrients applied	208.51	63.80	275.69	2,293.54
Anticipated crop nutrient removal	240.00	45.00	198.00	0.00
Actual crop nutrient removal	347.21	49.74	210.44	1,165.96
Nutrient balance	-138.70	14.06	65.25	1,127.57
Applied to removed ratio	0.60	1.28	1.31	1.97

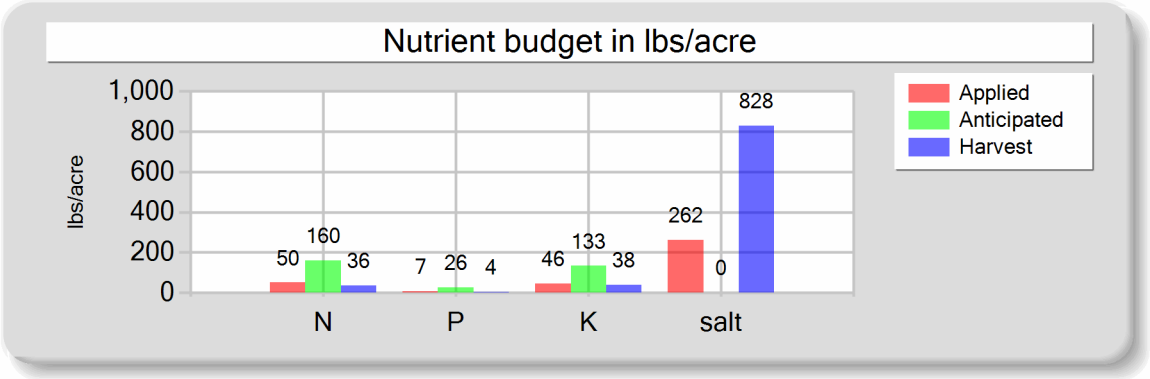
Fresh water applied	
48,815,700.00	*gallons*
1,797.72	*acre-inches*
31.00	*inches/acre*

Process wastewater applied	
12,282,660.00	*gallons*
452.33	*acre-inches*
7.80	*inches/acre*

Total harvests for the crop	
1	*harvests*

6B - 11/01/2022: Oats, silage-soft dough

Field name: 6B | Crop: Oats, silage-soft dough | Plant date: 11/01/2022



	Total N (lbs/acre)	Total P (lbs/acre)	Total K (lbs/acre)	Total salt (lbs/acre)
Existing soil nutrient content	0.00	0.00	0.00	0.00
Plowdown credit	0.00	0.00	0.00	0.00
Commercial fertilizer / Other	0.00	0.00	0.00	0.00
Dry manure	0.00	0.00	0.00	0.00
Process wastewater	43.17	7.29	46.36	261.87
Fresh water	0.00	0.00	0.00	0.00
Atmospheric deposition	7.00	0.00	0.00	0.00
Total nutrients applied	50.17	7.29	46.36	261.87
Anticipated crop nutrient removal	160.00	25.60	132.80	0.00
Actual crop nutrient removal	36.21	4.08	37.76	828.12
Nutrient balance	13.96	3.21	8.60	-566.26
Applied to removed ratio	1.39	1.78	1.23	0.32

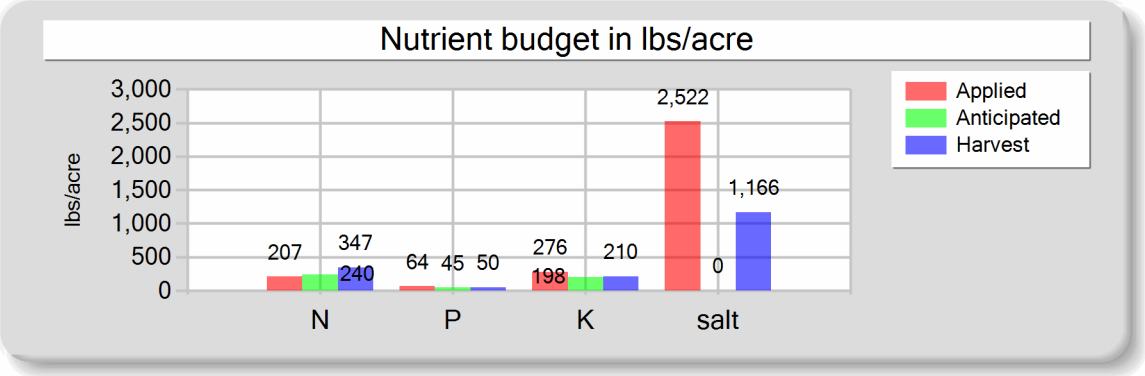
Fresh water applied
0.00 *gallons*
0.00 *acre-inches*
0.00 *inches/acre*

Process wastewater applied
59,368.00 *gallons*
2.19 *acre-inches*
0.31 *inches/acre*

Total harvests for the crop
1 *harvests*

6B - 05/15/2023: Corn, silage

Field name: 6B | Crop: Corn, silage | Plant date: 05/15/2023



	Total N (lbs/acre)	Total P (lbs/acre)	Total K (lbs/acre)	Total salt (lbs/acre)
Existing soil nutrient content	0.00	0.00	0.00	0.00
Plowdown credit	0.00	0.00	0.00	0.00
Commercial fertilizer / Other	0.00	0.00	0.00	0.00
Dry manure	0.00	0.00	0.00	0.00
Process wastewater	197.93	63.80	275.69	1,696.53
Fresh water	2.14	0.00	0.00	825.27
Atmospheric deposition	7.00	0.00	0.00	0.00
Total nutrients applied	207.07	63.80	275.69	2,521.80
Anticipated crop nutrient removal	240.00	45.00	198.00	0.00
Actual crop nutrient removal	347.21	49.74	210.44	1,165.96
Nutrient balance	-140.14	14.06	65.25	1,355.84
Applied to removed ratio	0.60	1.28	1.31	2.16

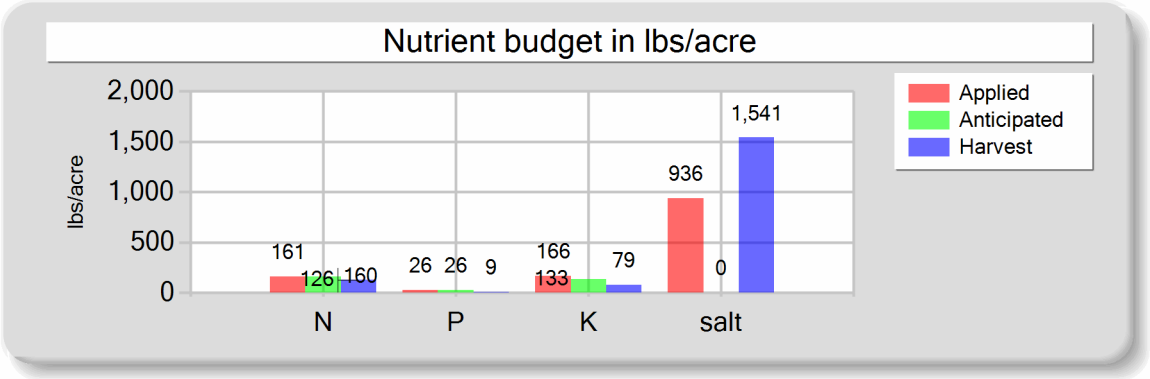
Fresh water applied	
5,891,544.00	*gallons*
216.97	*acre-inches*
31.00	*inches/acre*

Process wastewater applied	
1,482,390.00	*gallons*
54.59	*acre-inches*
7.80	*inches/acre*

Total harvests for the crop	
1	*harvests*

7 - 11/01/2022: Oats, silage-soft dough

Field name: 7 Crop: Oats, silage-soft dough Plant date: 11/01/2022



	Total N (lbs/acre)	Total P (lbs/acre)	Total K (lbs/acre)	Total salt (lbs/acre)
Existing soil nutrient content	0.00	0.00	0.00	0.00
Plowdown credit	0.00	0.00	0.00	0.00
Commercial fertilizer / Other	0.00	0.00	0.00	0.00
Dry manure	0.00	0.00	0.00	0.00
Process wastewater	154.36	26.06	165.74	936.26
Fresh water	0.00	0.00	0.00	0.00
Atmospheric deposition	7.00	0.00	0.00	0.00
Total nutrients applied	161.36	26.06	165.74	936.26
Anticipated crop nutrient removal	160.00	25.60	132.80	0.00
Actual crop nutrient removal	125.53	8.66	79.12	1,540.62
Nutrient balance	35.82	17.40	86.62	-604.36
Applied to removed ratio	1.29	3.01	2.09	0.61

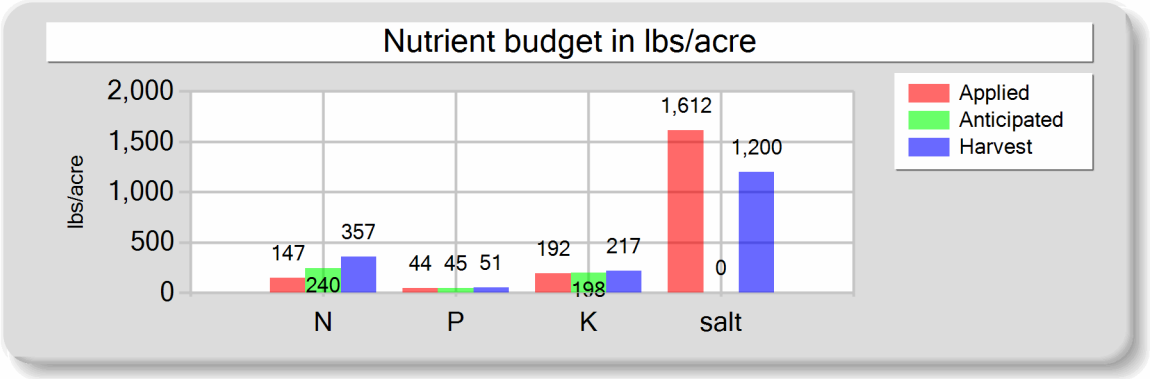
Fresh water applied
0.00 *gallons*
0.00 *acre-inches*
0.00 *inches/acre*

Process wastewater applied
940,000.00 *gallons*
34.62 *acre-inches*
1.12 *inches/acre*

Total harvests for the crop
1 *harvests*

7 - 05/15/2023: Corn, silage

Field name: 7 Crop: Corn, silage Plant date: 05/15/2023



	Total N (lbs/acre)	Total P (lbs/acre)	Total K (lbs/acre)	Total salt (lbs/acre)
Existing soil nutrient content	0.00	0.00	0.00	0.00
Plowdown credit	0.00	0.00	0.00	0.00
Commercial fertilizer / Other	0.00	0.00	0.00	0.00
Dry manure	0.00	0.00	0.00	0.00
Process wastewater	137.63	44.36	191.70	1,179.68
Fresh water	2.10	0.00	0.00	431.95
Atmospheric deposition	7.00	0.00	0.00	0.00
Total nutrients applied	146.73	44.36	191.70	1,611.63
Anticipated crop nutrient removal	240.00	45.00	198.00	0.00
Actual crop nutrient removal	357.42	51.20	216.63	1,200.26
Nutrient balance	-210.69	-6.84	-24.93	411.37
Applied to removed ratio	0.41	0.87	0.88	1.34

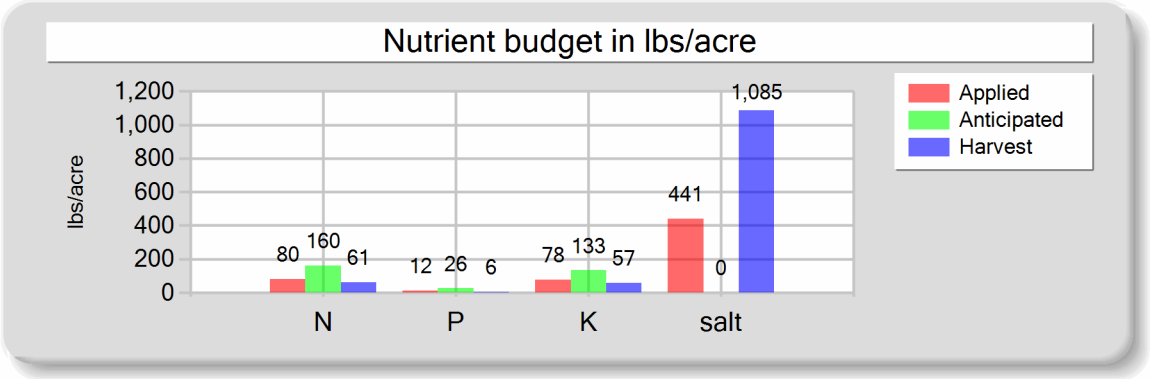
Fresh water applied	
26,091,150.00	*gallons*
960.85	*acre-inches*
31.00	*inches/acre*

Process wastewater applied	
4,564,870.00	*gallons*
168.11	*acre-inches*
5.42	*inches/acre*

Total harvests for the crop	
1	*harvests*

8 - 11/01/2022: Oats, silage-soft dough

Field name: 8 Crop: Oats, silage-soft dough Plant date: 11/01/2022



	Total N (lbs/acre)	Total P (lbs/acre)	Total K (lbs/acre)	Total salt (lbs/acre)
Existing soil nutrient content	0.00	0.00	0.00	0.00
Plowdown credit	0.00	0.00	0.00	0.00
Commercial fertilizer / Other	0.00	0.00	0.00	0.00
Dry manure	0.00	0.00	0.00	0.00
Process wastewater	72.72	12.28	78.09	441.09
Fresh water	0.00	0.00	0.00	0.00
Atmospheric deposition	7.00	0.00	0.00	0.00
Total nutrients applied	79.72	12.28	78.09	441.09
Anticipated crop nutrient removal	160.00	25.60	132.80	0.00
Actual crop nutrient removal	60.75	5.60	57.15	1,085.24
Nutrient balance	18.98	6.68	20.93	-644.15
Applied to removed ratio	1.31	2.19	1.37	0.41

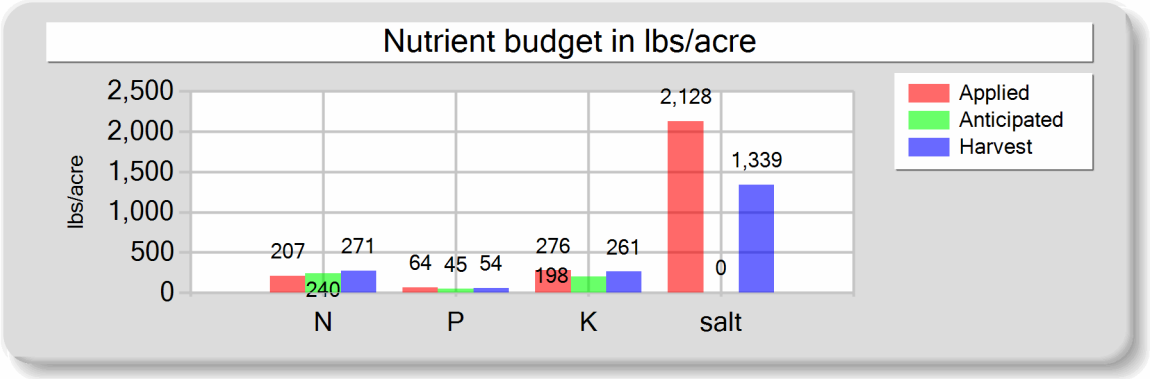
Fresh water applied	
0.00	*gallons*
0.00	*acre-inches*
0.00	*inches/acre*

Process wastewater applied	
200,000.00	*gallons*
7.37	*acre-inches*
0.53	*inches/acre*

Total harvests for the crop	
1	*harvests*

8 - 05/15/2023: Corn, silage

Field name: 8 Crop: Corn, silage Plant date: 05/15/2023



	Total N (lbs/acre)	Total P (lbs/acre)	Total K (lbs/acre)	Total salt (lbs/acre)
Existing soil nutrient content	0.00	0.00	0.00	0.00
Plowdown credit	0.00	0.00	0.00	0.00
Commercial fertilizer / Other	0.00	0.00	0.00	0.00
Dry manure	0.00	0.00	0.00	0.00
Process wastewater	197.93	63.80	275.69	1,696.53
Fresh water	2.06	0.00	0.00	431.95
Atmospheric deposition	7.00	0.00	0.00	0.00
Total nutrients applied	206.99	63.80	275.69	2,128.48
Anticipated crop nutrient removal	240.00	45.00	198.00	0.00
Actual crop nutrient removal	271.43	54.29	261.07	1,338.51
Nutrient balance	-64.44	9.51	14.61	789.97
Applied to removed ratio	0.76	1.18	1.06	1.59

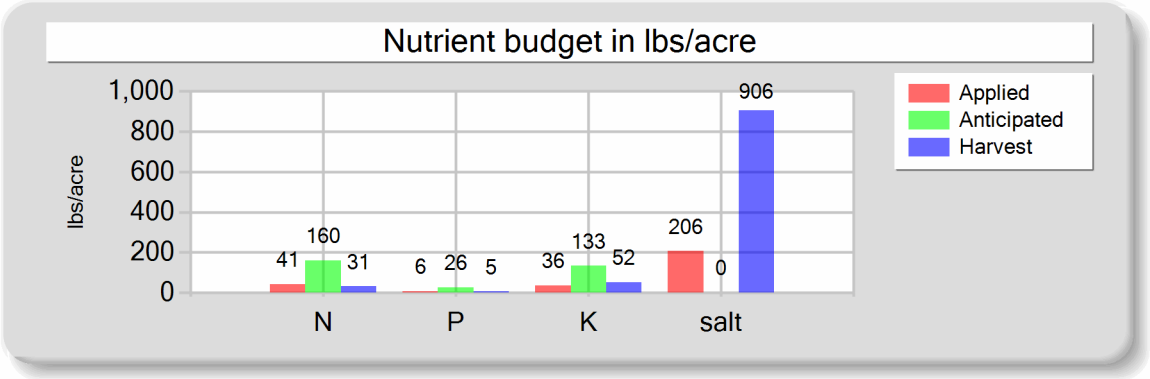
Fresh water applied	
11,783,100.00	*gallons*
433.93	*acre-inches*
31.00	*inches/acre*

Process wastewater applied	
2,964,780.00	*gallons*
109.18	*acre-inches*
7.80	*inches/acre*

Total harvests for the crop	
1	*harvests*

9 - 11/01/2022: Oats, silage-soft dough

Field name: 9 Crop: Oats, silage-soft dough Plant date: 11/01/2022



	Total N (lbs/acre)	Total P (lbs/acre)	Total K (lbs/acre)	Total salt (lbs/acre)
Existing soil nutrient content	0.00	0.00	0.00	0.00
Plowdown credit	0.00	0.00	0.00	0.00
Commercial fertilizer / Other	0.00	0.00	0.00	0.00
Dry manure	0.00	0.00	0.00	0.00
Process wastewater	33.94	5.73	36.44	205.84
Fresh water	0.00	0.00	0.00	0.00
Atmospheric deposition	7.00	0.00	0.00	0.00
Total nutrients applied	40.94	5.73	36.44	205.84
Anticipated crop nutrient removal	160.00	25.60	132.80	0.00
Actual crop nutrient removal	30.79	5.22	51.93	906.02
Nutrient balance	10.15	0.51	-15.49	-700.17
Applied to removed ratio	1.33	1.10	0.70	0.23

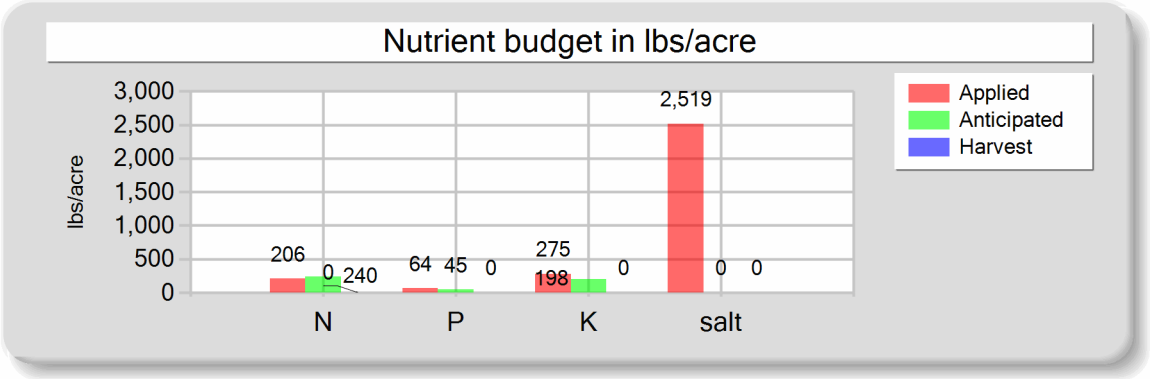
Fresh water applied
0.00 *gallons*
0.00 *acre-inches*
0.00 *inches/acre*

Process wastewater applied
120,000.00 *gallons*
4.42 *acre-inches*
0.25 *inches/acre*

Total harvests for the crop
1 *harvests*

9 - 05/15/2023: Corn, silage

Field name: 9 Crop: Corn, silage Plant date: 05/15/2023



	Total N (lbs/acre)	Total P (lbs/acre)	Total K (lbs/acre)	Total salt (lbs/acre)
Existing soil nutrient content	0.00	0.00	0.00	0.00
Plowdown credit	0.00	0.00	0.00	0.00
Commercial fertilizer / Other	0.00	0.00	0.00	0.00
Dry manure	0.00	0.00	0.00	0.00
Process wastewater	197.63	63.70	275.27	1,693.97
Fresh water	1.79	0.00	0.00	825.29
Atmospheric deposition	7.00	0.00	0.00	0.00
Total nutrients applied	206.42	63.70	275.27	2,519.26
Anticipated crop nutrient removal	240.00	45.00	198.00	0.00
Actual crop nutrient removal	0.00	0.00	0.00	0.00
Nutrient balance	206.42	63.70	275.27	2,519.26
Applied to removed ratio	0.00	0.00	0.00	0.00

Fresh water applied	
15,150,060.00	*gallons*
557.93	*acre-inches*
31.00	*inches/acre*

Process wastewater applied	
3,806,100.00	*gallons*
140.17	*acre-inches*
7.79	*inches/acre*

Total harvests for the crop	
1	*harvests*

NUTRIENT ANALYSES

A. MANURE ANALYSES

Manure 2091363-01

Sample and source description: Manure 2091363-01

Sample date: 09/13/2022 Material type: Corral solids Source of analysis: Lab analysis Method of reporting: Dry-weight

Moisture: 14.0 %

	Total N (mg/kg)	Total P (mg/kg)	Total K (mg/kg)	Calcium (mg/kg)	Magnesium (mg/kg)	Sodium (mg/kg)	Sulfur (mg/kg)	Chloride (mg/kg)	Total salt (mg/kg)	TFS (%)
Value	3,930.00	8,490.00	14,600.00	15,000.00	7,400.00	4,200.00	5,320.00	7,010.00		34.50
DL	0.50	0.50	0.50	5.00	5.00	5.00	1.00	0.50		0.10

B. PROCESS WASTEWATER ANALYSES

Lagoon Water 2302049-01

Sample and source description: Lagoon Water 2302049-01

Sample date: 01/31/2023 Material type: Process wastewater Source of analysis: Lab analysis pH:

	Kjeldahl-N (mg/L)	NH4-N (mg/L)	NH3-N (mg/L)	Nitrate-N (mg/L)	Total P (mg/L)	Total K (mg/L)	Calcium (mg/L)	Magnes. (mg/L)	Sodium (mg/L)	Bicarb. (mg/L)	Carb. (mg/L)	Sulfate (mg/L)	Chloride (mg/L)	EC (µmhos/cm)	TDS (mg/L)
Value	610.00	610.00			103.00	655.00								4,860.00	3,700
DL	0.60	0.60			0.10	10.00								1.00	1

Lagoon Water 2303388-01

Sample and source description: Lagoon Water 2303388-01

Sample date: 03/27/2023 Material type: Process wastewater Source of analysis: Lab analysis pH:

	Kjeldahl-N (mg/L)	NH4-N (mg/L)	NH3-N (mg/L)	Nitrate-N (mg/L)	Total P (mg/L)	Total K (mg/L)	Calcium (mg/L)	Magnes. (mg/L)	Sodium (mg/L)	Bicarb. (mg/L)	Carb. (mg/L)	Sulfate (mg/L)	Chloride (mg/L)	EC (µmhos/cm)	TDS (mg/L)
Value	421.00	421.00			90.20	775.00								4,940.00	3,000
DL	0.60	10.00			0.10	10.00								1.00	1

Lagoon Water 2307240-01

Sample and source description: Lagoon Water 2307240-01

Sample date: 07/17/2023 Material type: Process wastewater Source of analysis: Lab analysis pH:

	Kjeldahl-N (mg/L)	NH4-N (mg/L)	NH3-N (mg/L)	Nitrate-N (mg/L)	Total P (mg/L)	Total K (mg/L)	Calcium (mg/L)	Magnes. (mg/L)	Sodium (mg/L)	Bicarb. (mg/L)	Carb. (mg/L)	Sulfate (mg/L)	Chloride (mg/L)	EC (µmhos/cm)	TDS (mg/L)
Value	112.00	103.00			36.10	156.00								1,940.00	960
DL	0.60	0.60			0.10	10.00								1.00	1

Lagoon Water 2310026-01

Sample and source description: Lagoon Water 2310026-01

Sample date: 09/30/2023 Material type: Process wastewater Source of analysis: Lab analysis pH:

	Kjeldahl-N (mg/L)	NH4-N (mg/L)	NH3-N (mg/L)	Nitrate-N (mg/L)	Total P (mg/L)	Total K (mg/L)	Calcium (mg/L)	Magnes. (mg/L)	Sodium (mg/L)	Bicarb. (mg/L)	Carb. (mg/L)	Sulfate (mg/L)	Chloride (mg/L)	EC (µmhos/cm)	TDS (mg/L)
Value	276.00	234.00			65.20	416.00								3,920.00	2,500
DL	0.60	0.60			0.10	10.00								1.00	1

C. FRESH WATER ANALYSES

Ag Well #1

2312446-02

Sample description: 2312446-02

Sample date: 12/22/2023 Source of analysis: Lab analysis

	Total N (mg/L)	NH4-N (mg/L)	Nitrate-N (mg/L)	Calcium (mg/L)	Magnesium (mg/L)	Sodium (mg/L)	Bicarbonate (mg/L)	Carbonate (mg/L)	Sulfate (mg/L)	Chloride (mg/L)	EC (µmhos/cm)	TDS (mg/L)
Value	62.20										1,550.00	20,000
DL	1.40										1.00	1

Ag Well #3

Ag Well #3

2312446-03

Sample description: 2312446-03

Sample date: 12/22/2023 Source of analysis: Lab analysis

	Total N (mg/L)	NH4-N (mg/L)	Nitrate-N (mg/L)	Calcium (mg/L)	Magnesium (mg/L)	Sodium (mg/L)	Bicarbonate (mg/L)	Carbonate (mg/L)	Sulfate (mg/L)	Chloride (mg/L)	EC (µmhos/cm)	TDS (mg/L)
Value	27.80										1,060.00	660
DL	1.40										1.00	1

Ag Well #4

2312446-04

Sample description: 2312446-04

Sample date: 12/22/2023 Source of analysis: Lab analysis

	Total N (mg/L)	NH4-N (mg/L)	Nitrate-N (mg/L)	Calcium (mg/L)	Magnesium (mg/L)	Sodium (mg/L)	Bicarbonate (mg/L)	Carbonate (mg/L)	Sulfate (mg/L)	Chloride (mg/L)	EC (µmhos/cm)	TDS (mg/L)
Value											263.00	150
DL											1.00	1

Ag Well #5

2312446-05

Sample description: 2312446-05

Sample date: 12/22/2023 Source of analysis: Lab analysis

	Total N (mg/L)	NH4-N (mg/L)	Nitrate-N (mg/L)	Calcium (mg/L)	Magnesium (mg/L)	Sodium (mg/L)	Bicarbonate (mg/L)	Carbonate (mg/L)	Sulfate (mg/L)	Chloride (mg/L)	EC (µmhos/cm)	TDS (mg/L)
Value											216.00	38
DL											1.00	1

Canal Water

Canal Water

2310024-01

Sample description: 2310024-01

Sample date: 09/29/2023 Source of analysis: Lab analysis

	Total N (mg/L)	NH4-N (mg/L)	Nitrate-N (mg/L)	Calcium (mg/L)	Magnesium (mg/L)	Sodium (mg/L)	Bicarbonate (mg/L)	Carbonate (mg/L)	Sulfate (mg/L)	Chloride (mg/L)	EC (µmhos/cm)	TDS (mg/L)
Value	0.51										39.10	85
DL	0.40										1.00	1

Domestic Well

2312446-01

Sample description: 2312446-01

Sample date: 12/22/2023 Source of analysis: Lab analysis

	Total N (mg/L)	NH4-N (mg/L)	Nitrate-N (mg/L)	Calcium (mg/L)	Magnesium (mg/L)	Sodium (mg/L)	Bicarbonate (mg/L)	Carbonate (mg/L)	Sulfate (mg/L)	Chloride (mg/L)	EC (µmhos/cm)	TDS (mg/L)
Value	18.20										941.00	
DL	0.40										1.00	

D. SOIL ANALYSES

No soil analyses entered.

E. PLANT TISSUE ANALYSES

1 - 05/15/2023: Corn, silage

1 - 05/15/2023: Corn, silage

2310206-01

Sample and source description: 2310206-01

Sample date: 10/08/2023 Source of analysis: Lab analysis Method of reporting: Dry-weight

Moisture: 68.3 %

	Total N (mg/kg)	Total P (mg/kg)	Total K (mg/kg)	Total salt (mg/kg)	TFS (%)
Value	17,600.00	2,140.00	8,790.00		5.62
DL	0.50	0.50	0.50		0.10

2A - 05/15/2023: Corn, silage

2310206-02

Sample and source description: 2310206-02

Sample date: 10/08/2023 Source of analysis: Lab analysis Method of reporting: Dry-weight

Moisture: 70.1 %

	Total N (mg/kg)	Total P (mg/kg)	Total K (mg/kg)	Total salt (mg/kg)	TFS (%)
Value	16,200.00	1,740.00	9,380.00		5.68
DL	0.50	0.50	0.50		0.10

2B - 05/15/2023: Corn, silage

2310206-02

Sample and source description: 2310206-02

Sample date: 10/08/2023 Source of analysis: Lab analysis Method of reporting: Dry-weight

Moisture: 70.1 %

	Total N (mg/kg)	Total P (mg/kg)	Total K (mg/kg)	Total salt (mg/kg)	TFS (%)
Value	16,200.00	1,740.00	9,380.00		5.68
DL	0.50	0.50	0.50		0.10

3 - 11/01/2022: Oats, silage-soft dough

2305066-03

Sample and source description: 2305066-03

Sample date: 05/02/2023 Source of analysis: Lab analysis Method of reporting: Dry-weight

Moisture: 67.8 %

	Total N (mg/kg)	Total P (mg/kg)	Total K (mg/kg)	Total salt (mg/kg)	TFS (%)
Value	13,200.00	1,060.00	9,390.00		11.60
DL	2.50	0.50	0.50		0.10

3 - 05/15/2023: Corn, silage

2310206-03

Sample and source description: 2310206-03

Sample date: 10/08/2023 Source of analysis: Lab analysis Method of reporting: Dry-weight

Moisture: 68.2 %

	Total N (mg/kg)	Total P (mg/kg)	Total K (mg/kg)	Total salt (mg/kg)	TFS (%)
Value	16,100.00	1,610.00	8,940.00		4.83
DL	0.50	0.50	0.50		0.10

4 - 11/01/2022: Oats, silage-soft dough

2305066-04

Sample and source description: 2305066-04

Sample date: 05/02/2023 Source of analysis: Lab analysis Method of reporting: Dry-weight

Moisture: 57.6 %

	Total N (mg/kg)	Total P (mg/kg)	Total K (mg/kg)	Total salt (mg/kg)	TFS (%)
Value	7,210.00	1,230.00	12,200.00		12.40
DL	2.50	0.50	0.50		0.10

4 - 05/15/2023: Corn, silage

2310206-04

Sample and source description: 2310206-04

Sample date: 10/08/2023 Source of analysis: Lab analysis Method of reporting: Dry-weight

Moisture: 69.5 %

	Total N (mg/kg)	Total P (mg/kg)	Total K (mg/kg)	Total salt (mg/kg)	TFS (%)
Value	21,500.00	2,410.00	12,400.00		7.84
DL	0.50	0.50	0.50		0.10

5A - 11/01/2022: Oats, silage-soft dough

2305066-05

Sample and source description: 2305066-05

Sample date: 05/02/2023 Source of analysis: Lab analysis Method of reporting: Dry-weight

Moisture: 81.7 %

	Total N (mg/kg)	Total P (mg/kg)	Total K (mg/kg)	Total salt (mg/kg)	TFS (%)
Value	6,890.00	625.00	5,990.00		14.30
DL	2.50	0.50	0.50		0.10

5A - 05/15/2023: Corn, silage

2310206--05

Sample and source description: 2310206--05

Sample date: 10/08/2023 Source of analysis: Lab analysis Method of reporting: Dry-weight

Moisture: 68.9 %

	Total N (mg/kg)	Total P (mg/kg)	Total K (mg/kg)	Total salt (mg/kg)	TFS (%)
Value	12,400.00	1,640.00	9,920.00		5.28
DL	0.50	0.50	0.50		0.10

5B - 11/01/2022: Oats, silage-soft dough

2305066-05

Sample and source description: 2305066-05

Sample date: 05/02/2023 Source of analysis: Lab analysis Method of reporting: Dry-weight

Moisture: 81.7 %

	Total N (mg/kg)	Total P (mg/kg)	Total K (mg/kg)	Total salt (mg/kg)	TFS (%)
Value	6,890.00	625.00	5,990.00		14.30
DL	2.50	0.50	0.50		0.10

5B - 05/15/2023: Corn, silage

2310206-05

Sample and source description: 2310206-05

Sample date: 10/08/2023 Source of analysis: Lab analysis Method of reporting: Dry-weight

Moisture: 68.9 %

	Total N (mg/kg)	Total P (mg/kg)	Total K (mg/kg)	Total salt (mg/kg)	TFS (%)
Value	12,400.00	1,640.00	9,920.00		5.28
DL	0.50	0.50	0.50		0.10

6A - 11/01/2022: Oats, silage-soft dough

2305066-06

Sample and source description: 2305066-06

Sample date: 05/02/2023 Source of analysis: Lab analysis Method of reporting: Dry-weight

Moisture: 75.8 %

	Total N (mg/kg)	Total P (mg/kg)	Total K (mg/kg)	Total salt (mg/kg)	TFS (%)
Value	5,160.00	582.00	5,380.00		11.80
DL	2.50	0.50	0.50		0.10

6A - 05/15/2023: Corn, silage

2310206-06

Sample and source description: 2310206-06

Sample date: 10/08/2023 Source of analysis: Lab analysis Method of reporting: Dry-weight

Moisture: 65.5 %

	Total N (mg/kg)	Total P (mg/kg)	Total K (mg/kg)	Total salt (mg/kg)	TFS (%)
Value	14,800.00	2,120.00	8,970.00		4.97
DL	0.50	0.50	0.50		0.10

6B - 11/01/2022: Oats, silage-soft dough

2305066-06

Sample and source description: 2305066-06

Sample date: 05/02/2023 Source of analysis: Lab analysis Method of reporting: Dry-weight

Moisture: 75.8 %

	Total N (mg/kg)	Total P (mg/kg)	Total K (mg/kg)	Total salt (mg/kg)	TFS (%)
Value	5,160.00	582.00	5,380.00		11.80
DL	2.50	0.50	0.50		0.10

6B - 05/15/2023: Corn, silage

2310206-06

Sample and source description: 2310206-06

Sample date: 10/08/2023 Source of analysis: Lab analysis Method of reporting: Dry-weight

Moisture: 65.5 %

	Total N (mg/kg)	Total P (mg/kg)	Total K (mg/kg)	Total salt (mg/kg)	TFS (%)
Value	14,800.00	2,120.00	8,970.00		4.97
DL	0.50	0.50	0.50		0.10

7 - 11/01/2022: Oats, silage-soft dough

2305066-07

Sample and source description: 2305066-07

Sample date: 05/02/2023 Source of analysis: Lab analysis Method of reporting: Dry-weight

Moisture: 68.3 %

	Total N (mg/kg)	Total P (mg/kg)	Total K (mg/kg)	Total salt (mg/kg)	TFS (%)
Value	13,200.00	911.00	8,320.00		16.20
DL	2.50	0.50	0.50		0.10

7 - 05/15/2023: Corn, silage

2310206-07

Sample and source description: 2310206-07

Sample date: 10/08/2023 Source of analysis: Lab analysis Method of reporting: Dry-weight

Moisture: 59.5 %

	Total N (mg/kg)	Total P (mg/kg)	Total K (mg/kg)	Total salt (mg/kg)	TFS (%)
Value	13,400.00	1,210.00	7,930.00		13.80
DL	0.50	0.50	0.50		0.10

8 - 11/01/2022: Oats, silage-soft dough

2305066-08

Sample and source description: 2305066-08

Sample date: 05/02/2023 Source of analysis: Lab analysis Method of reporting: Dry-weight

Moisture: 75.7 %

	Total N (mg/kg)	Total P (mg/kg)	Total K (mg/kg)	Total salt (mg/kg)	TFS (%)
Value	8,620.00	794.00	8,110.00		15.40
DL	2.50	0.50	0.50		0.10

8 - 05/15/2023: Corn, silage

2310206-08

Sample and source description: 2310206-08

Sample date: 10/08/2023 Source of analysis: Lab analysis Method of reporting: Dry-weight

Moisture: 70.4 %

	Total N (mg/kg)	Total P (mg/kg)	Total K (mg/kg)	Total salt (mg/kg)	TFS (%)
Value	13,100.00	2,620.00	12,600.00		6.46
DL	0.50	0.50	0.50		0.10

9 - 11/01/2022: Oats, silage-soft dough

2305067-01

Sample and source description: 2305067-01

Sample date: 05/02/2023 Source of analysis: Lab analysis Method of reporting: Dry-weight

Moisture: 74.6 %

	Total N (mg/kg)	Total P (mg/kg)	Total K (mg/kg)	Total salt (mg/kg)	TFS (%)
Value	4,180.00	709.00	70,500.00		12.30
DL	5.00	0.50	0.50		0.10

F. SUBSURFACE (TILE) DRAINAGE ANALYSES

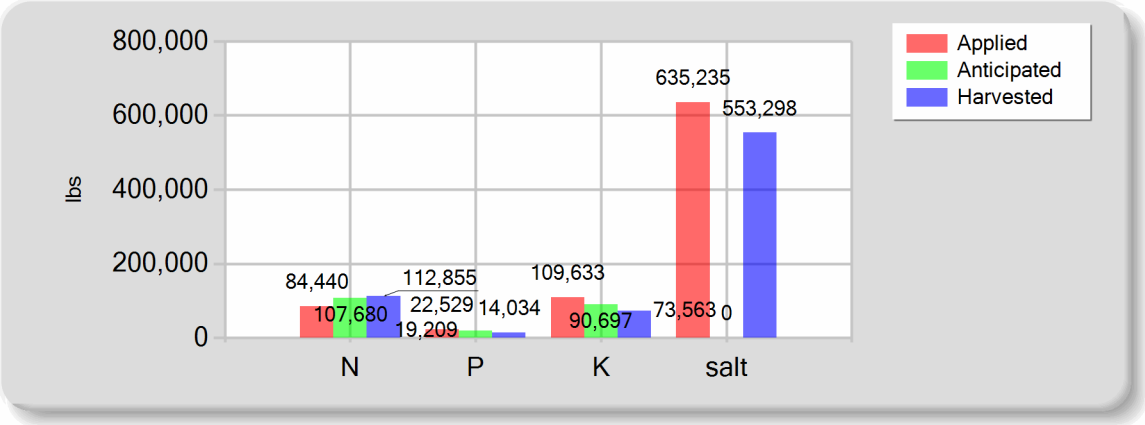
No subsurface (tile) drainage analyses entered.

NUTRIENT APPLICATIONS, POTENTIAL REMOVAL, AND BALANCE

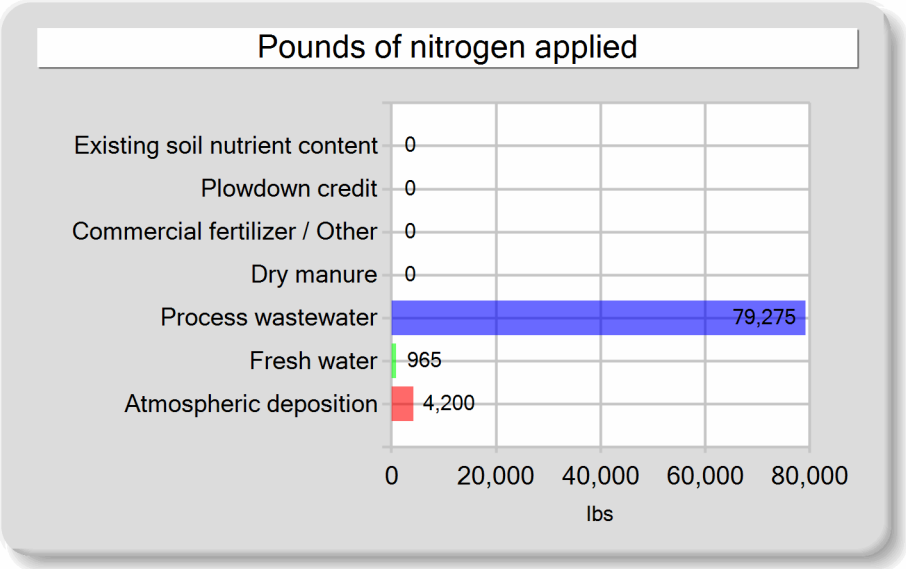
A. SUMMARY OF NUTRIENT APPLICATIONS, POTENTIAL REMOVAL, AND BALANCE

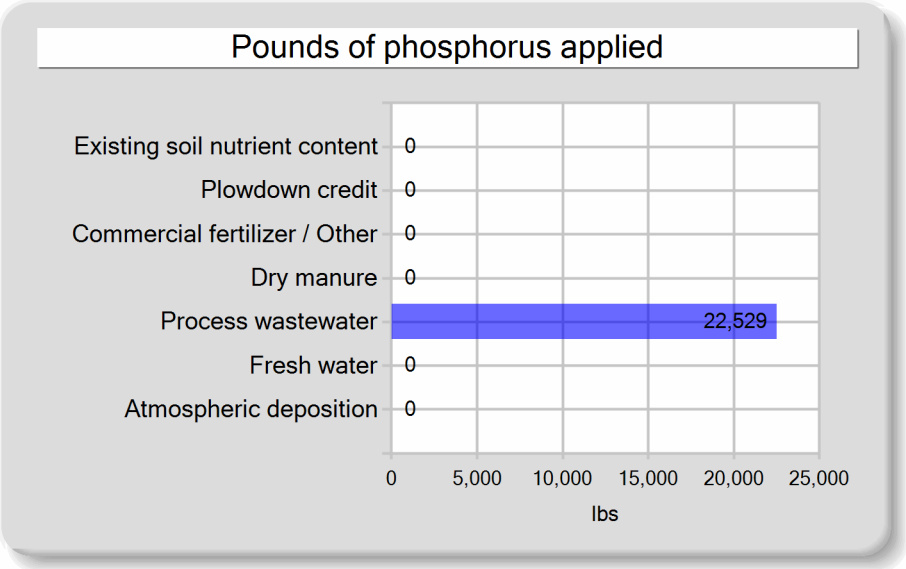
	Total N (lbs)	Total P (lbs)	Total K (lbs)	Total salt (lbs)
Existing soil nutrient content	0.00	0.00	0.00	0.00
Plowdown credit	0.00	0.00	0.00	0.00
Commercial fertilizer / Other	0.00	0.00	0.00	0.00
Dry manure	0.00	0.00	0.00	0.00
Process wastewater	79,274.99	22,529.00	109,633.17	457,853.93
Fresh water	965.16	0.00	0.00	177,380.59
Atmospheric deposition	4,200.00	0.00	0.00	0.00
Total nutrients applied	84,440.15	22,529.00	109,633.17	635,234.51
Anticipated crop nutrient removal	107,680.00	19,208.80	90,697.40	0.00
Actual crop nutrient removal	112,854.95	14,034.28	73,562.77	553,297.90
Nutrient balance	-28,414.80	8,494.71	36,070.40	81,936.61
Applied to removed ratio	0.75	1.61	1.49	1.15

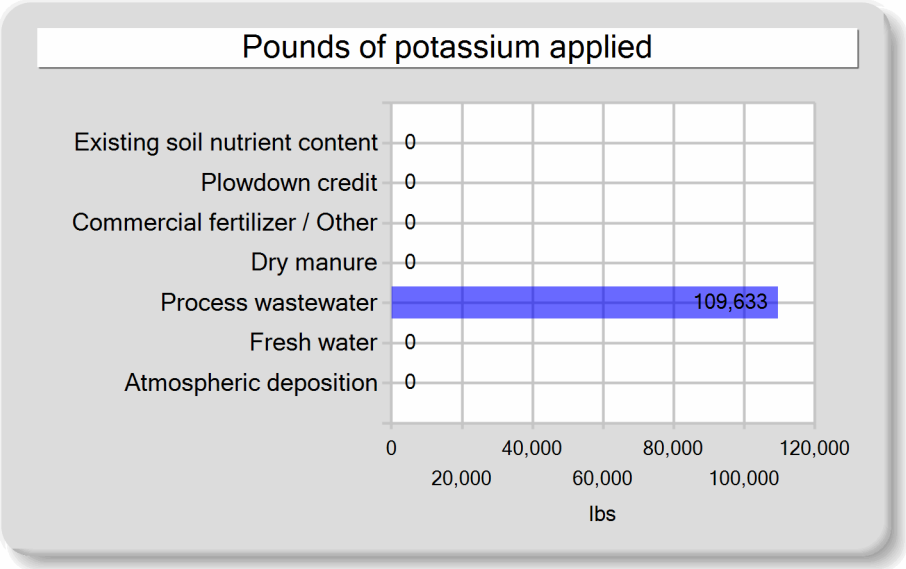
B. POUNDS OF NUTRIENT APPLIED VS. CROP REMOVAL

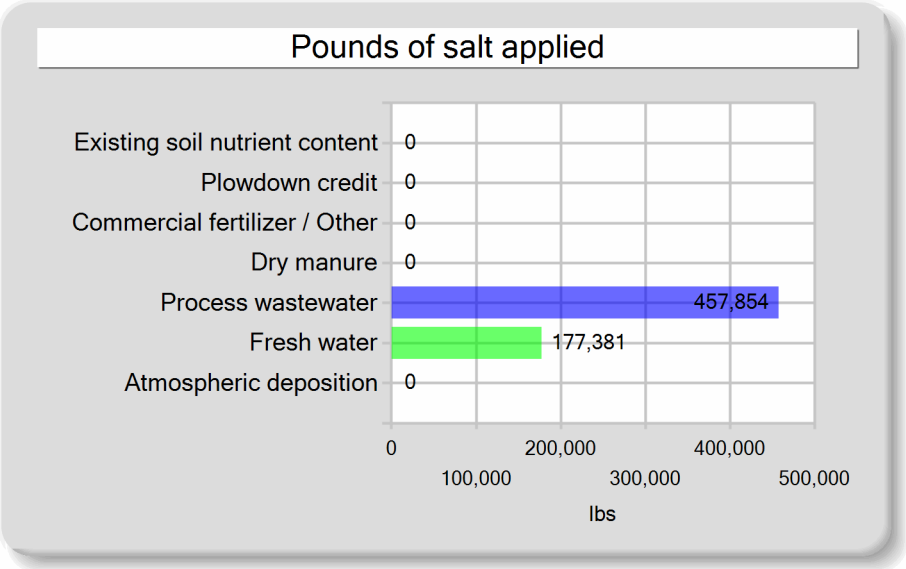


C. POUNDS OF NUTRIENT APPLIED BY MATERIAL TYPE









EXCEPTION REPORTING

A. MANURE, PROCESS WASTEWATER, AND OTHER DAIRY WASTE DISCHARGES

The following is a summary of all manure and process wastewater discharges from the production area to surface water or to land areas (land application areas or otherwise) when not in accordance with the facility's Nutrient Management Plan.

No manure or process wastewater discharges occurred during the reporting period.

B. STORM WATER DISCHARGES

The following is a summary of all storm water discharges from the production area to surface water during the reporting period when not in accordance with the facility 's Nutrient Management Plan.

No stormwater discharges occurred during the reporting period.

C. LAND APPLICATION AREA TO SURFACE WATER DISCHARGES

The following is a summary of all discharges from the land application area to surface water that have occurred during the reporting period when not in accordance with the facility's Nutrient Management Plan.

No land application area to surface water discharges occurred during the reporting period.

NUTRIENT MANAGEMENT PLAN AND EXPORT AGREEMENT STATEMENTS

A. NUTRIENT MANAGEMENT PLAN STATEMENTS

Was the facility's NMP updated in the reporting period?	No
Was the facility's NMP developed by a certified nutrient management planner (specialist) as specified in Attachment C of the General Order?	Yes
Was the facility's NMP approved by a certified nutrient management planner (specialist) as specified in Attachment C of the General Order?	Yes

B. EXPORT AGREEMENT STATEMENT

Are there any written agreements with third parties to receive manure or process wastewater that are new or were revised within the reporting period?	No

ADDITIONAL NOTES

A. NOTES

*Field #9 Corn Silage was not sampled in 2023, we will be sure to sample in 2024.

CERTIFICATION

A. OWNER AND/OR OPERATOR CERTIFICATION

I certify under penalty of law that I have personally examined and am familiar with the information submitted in this document and all attachments and that, based on my inquiry of those individuals immediately responsible for obtaining the information, I believe that the information is true, accurate, and complete. I am aware that there are significant penalties for submitting false information, including the possibility of fine and imprisonment.

X	
SIGNATURE OF OWNER OF FACILITY	SIGNATURE OF OPERATOR OF FACILITY
George teVelde	SAME AS OWNER
PRINT OR TYPE NAME	PRINT OR TYPE NAME
7/12/24	7/12/24
DATE	DATE

ATTACHMENTS

A. REQUIRED ATTACHMENTS

The following lists the required documents that should be attached to the Annual Report when submitted .

Annual Dairy Facility Assessment	N/A
Provide an Annual Dairy Facility Assessment (an update to the Preliminary Dairy Facility Assessment in Attachment A) for each reporting period. On the PDFA Final page, click on the ADFA Report button to generate an ADFA report after updating information as needed .	

Manure/Process Wastewater Tracking Manifests	N/A
Provide copies of all manure/process wastewater tracking manifests for the reporting period, signed by both the owner/operator and the hauler.	

Corrective Actions Documents	N/A
Provide records documenting any corrective actions taken to correct deficiencies noted as a result of the inspections required in the Monitoring Requirements of the General Order. Deficiencies not corrected in 30 days must be accompanied by an explanation of the factors preventing immediate correction.	

Groundwater Monitoring	N/A
Dischargers that monitor supply wells or subsurface (tile) drainage systems, or that have monitoring well systems must submit monitoring results as directed in the General Order, Groundwater Reporting Section starting on page MRP-13.	

Storm Water Monitoring	N/A
Dischargers that are required to monitor storm water more frequently than required in the General Order must submit monitoring results as directed in the General Order, Storm Water Reporting Section on page MRP-14.	